

LECTURE NOTES ON INTEGRABLE SYSTEMS

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ABSTRACT. These notes are based on the course *Introduction to Integrable Systems*, delivered by Professor Liming Ling during the Fall semester of 2025–2026.

The full L^AT_EX source code of these notes is available at: <https://codeberg.org/kill11kill/courseNotes>.

Week 1: Algebro-Geometric Solutions for the mKdV Equation (1)

2025-09-16

Objectives: Derive the mKdV equation from the associated spectral problem and construct the AKNS hierarchy by explicit recursion.

The focusing modified Korteweg–de Vries (mKdV) equation is given by

$$(1.1) \quad u_t + 6u^2u_x + u_{xxx} = 0, \quad u(x, t) \in \mathbb{R}, \quad (x, t) \in \mathbb{R}^2.$$

A standard method for establishing integrability is via a Lax pair. We introduce a matrix function $\Phi(x, t)$ satisfying

$$\Phi_x = U\Phi, \quad \Phi_t = V\Phi,$$

where U and V are 2×2 matrices depending on u and its derivatives. The compatibility condition

$$\Phi_{xt} = \Phi_{tx} \iff U_t - V_x + [U, V] = 0,$$

with $[U, V] = UV - VU$, is equivalent to the zero-curvature equation.

Following the AKNS formalism, the spatial part of the Lax pair is taken as

$$(1.2) \quad \Phi_x = U\Phi, \quad U = -i\lambda\sigma_3 + Q,$$

where

$$(1.3) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad \sigma_3 = \text{diag}(1, -1).$$

We introduce the transformation

$$(1.4) \quad \Phi(x; \lambda) = m(x; \lambda)e^{-i\lambda x\sigma_3}, \quad m(x; \lambda) \rightarrow \mathbb{I} \text{ as } \lambda \rightarrow \infty.$$

Substituting into the spectral problem (1.2) gives

$$(1.5) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i\lambda \sigma_3 + Q.$$

Define

$$(1.6) \quad \Psi = m \sigma_3 m^{-1}.$$

Differentiating Ψ and using (1.5), we compute

$$\begin{aligned} \Psi_x &= m_x \sigma_3 m^{-1} - m \sigma_3 m^{-1} m_x m^{-1} \\ &= [m_x m^{-1}, m \sigma_3 m^{-1}] \\ &= [m_x m^{-1} - i\lambda m \sigma_3 m^{-1}, m \sigma_3 m^{-1}] \\ &= [U, \Psi]. \end{aligned}$$

Thus Ψ satisfies

$$(1.7) \quad \Psi_x = [U, \Psi],$$

which is the stationary zero-curvature equation.

The matrix $m(x; \lambda)$ admits the asymptotic expansion

$$(1.8) \quad m(x; \lambda) = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + \dots.$$

From (1.6) and (1.8) we obtain the formal expansion

$$(1.9) \quad \Psi = \Psi_0 + \Psi_1\lambda^{-1} + \Psi_2\lambda^{-2} + \dots, \quad \Psi_0 = \sigma_3.$$

Substituting this into (1.7) yields

$$\begin{aligned} \sum_{j=1}^{\infty} \Psi_{j,x} \lambda^{-j} &= -i\lambda \left[\sigma_3, \sum_{j=1}^{\infty} \Psi_j \lambda^{-j} \right] + \left[Q, \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right] \\ &= -i \sum_{j=0}^{\infty} \left[\sigma_3, \Psi_{j+1} \right] \lambda^{-j} + \sum_{j=0}^{\infty} \left[Q, \Psi_j \right] \lambda^{-j}. \end{aligned}$$

At order $O(1)$ we find

$$-i[\sigma_3, \Psi_1] + [Q, \sigma_3] = 0 \iff [i\Psi_1 + Q, \sigma_3] = 0,$$

which implies

$$\Psi_1^{\text{off}} = iQ.$$

For general $j \geq 1$, the coefficient of λ^{-j} gives

$$(1.10) \quad \Psi_{j,x} = -i[\sigma_3, \Psi_{j+1}] + [Q, \Psi_j] = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j].$$

Hence

$$(1.11) \quad \Psi_{j,x}^{\text{off}} = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j^{\text{diag}}] \iff \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}]),$$

and

$$(1.12) \quad \Psi_{j,x}^{\text{diag}} = [Q, \Psi_j^{\text{off}}] \iff \Psi_j^{\text{diag}} = \partial_x^{-1}([Q, \Psi_j^{\text{off}}]).$$

Thus, the recursion relation (1.11) provides a direct way to compute the off-diagonal coefficients algebraically in terms of lower-order terms, while the diagonal recursion (1.12) requires performing an integration. In practice, this makes the off-diagonal part more convenient to obtain explicitly, whereas the diagonal part is generally harder to compute in closed form.

By (1.6), the additional constraint $\Psi^2 = \mathbb{I}$ provides algebraic relations among the coefficients. Substituting the expansion (1.9) yields

$$\mathbb{I} = \left(\sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right)^2 = \sum_{j=0}^{\infty} \left(\sum_{k=0}^j \Psi_k \Psi_{j-k} \right) \lambda^{-j},$$

so that

$$(1.13) \quad \sum_{k=0}^j \Psi_k \Psi_{j-k} = 0, \quad j \geq 1.$$

Separating diagonal and off-diagonal parts, for $j \geq 1$ we obtain

$$(1.14) \quad \sum_{k=0}^j (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0,$$

$$(1.15) \quad \sum_{k=0}^j (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Since $\Psi_0 = \sigma_3$, the off-diagonal constraint (1.15) for $j \geq 2$ takes the form

$$\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Noting that $\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 = 0$, this simplifies to

$$(1.16) \quad \sum_{k=1}^{j-1} \left(\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}} \right) = 0.$$

It should be noted that from the off-diagonal constraint (1.15) one can only deduce the reduced form (1.16), which merely expresses a compatibility condition between lower-order coefficients. In particular, this relation does not provide a practical means to compute the diagonal parts explicitly.

For the diagonal part (1.14) we have

$$\sigma_3 \Psi_j^{\text{diag}} + \Psi_j^{\text{diag}} \sigma_3 + \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right) = 0, \quad j \geq 2.$$

Since $\sigma_3 \Psi_j^{\text{diag}} = \Psi_j^{\text{diag}} \sigma_3$, this simplifies to

$$(1.17) \quad \Psi_j^{\text{diag}} = -\frac{1}{2} \sigma_3 \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right), \quad j \geq 2.$$

We will use this reduced relation (1.17) to compute the diagonal part Ψ_j^{diag} recursively at each order. This provides an explicit algebraic formula for the diagonal coefficients in terms of lower-order diagonal and off-diagonal terms.

For $j = 1$, the condition (1.13) reads

$$\sigma_3 \Psi_1 + \Psi_1 \sigma_3 = 2\sigma_3 \Psi_1^{\text{diag}} = 0,$$

so that $\Psi_1 = \Psi_1^{\text{off}} = iQ$.

Using (1.11) and (1.17) recursively, we find

$$\Psi_2^{\text{off}} = -\frac{1}{2} \sigma_3 Q_x, \quad \Psi_2^{\text{diag}} = \frac{1}{2} \sigma_3 Q^2,$$

and hence

$$\Psi_2 = \frac{1}{2} \sigma_3 (Q^2 - Q_x).$$

For the next coefficient, we compute

$$\begin{aligned} \Psi_3^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{2,x}^{\text{off}} - [Q, \Psi_2^{\text{diag}}]) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - [Q, \frac{1}{2} \sigma_3 Q^2] \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - \frac{1}{2} (Q \sigma_3 Q^2 - \sigma_3 Q^2 Q) \right) \\ &= -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3. \end{aligned}$$

For the diagonal part,

$$\begin{aligned} \Psi_3^{\text{diag}} &= -\frac{1}{2} \sigma_3 \left(\Psi_1^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{off}} \Psi_1^{\text{off}} + \Psi_1^{\text{diag}} \Psi_2^{\text{diag}} + \Psi_2^{\text{diag}} \Psi_1^{\text{diag}} \right) \\ &= -\frac{1}{2} \sigma_3 \left((iQ) \left(-\frac{1}{2} \sigma_3 Q_x \right) + \left(-\frac{1}{2} \sigma_3 Q_x \right) (iQ) \right) \\ &= -\frac{1}{2} \sigma_3 \left(-\frac{i}{2} (Q \sigma_3 Q_x + \sigma_3 Q_x Q) \right) \\ &= \frac{i}{4} (Q_x Q - Q Q_x) = 0. \end{aligned}$$

Proceeding to Ψ_4 , the off-diagonal recursion (1.11) with $\Psi_3^{\text{diag}} = 0$ gives

$$\begin{aligned} \Psi_4^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{3,x}^{\text{off}} - [Q, \Psi_3^{\text{diag}}]) = \frac{i}{2} \sigma_3 \Psi_{3,x}^{\text{off}} \\ &= \frac{i}{2} \sigma_3 \partial_x \left(-\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3 \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{i}{4} Q_{xxx} + \frac{i}{2} (Q^3)_x \right) \\ &= \frac{1}{8} \sigma_3 Q_{xxx} - \frac{1}{4} \sigma_3 (Q^3)_x. \end{aligned}$$

Since $Q_x Q = Q Q_x$ for the present Q given in (1.3), we have

$$(Q^3)_x = Q_x Q^2 + Q Q_x Q + Q^2 Q_x = 3Q_x Q^2,$$

and therefore

$$\Psi_4^{\text{off}} = \frac{1}{8}\sigma_3 Q_{xxx} - \frac{3}{4}\sigma_3 Q_x Q^2.$$

For the diagonal part, only the nonvanishing combinations contribute; terms involving Ψ_1^{diag} or Ψ_3^{diag} are zero and omitted. Hence

$$\begin{aligned} \Psi_4^{\text{diag}} &= -\frac{1}{2}\sigma_3 \left(\Psi_1^{\text{off}} \Psi_3^{\text{off}} + \Psi_3^{\text{off}} \Psi_1^{\text{off}} + \Psi_2^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{diag}} \Psi_2^{\text{diag}} \right) \\ &= -\frac{1}{2}\sigma_3 \left((iQ) \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) + \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) (iQ) \right. \\ &\quad \left. + \left(-\frac{1}{2}\sigma_3 Q_x \right)^2 + \left(\frac{1}{2}\sigma_3 Q^2 \right)^2 \right) \\ &= -\frac{1}{8}\sigma_3 (Q Q_{xx} + Q_{xx} Q - 3Q^4 - Q_x^2). \end{aligned}$$

Since $(-i\lambda\Psi)_+ = -i\lambda\sigma_3 + Q$, we obtain

$$\Phi_x \Phi^{-1} = U = (-i\lambda\Psi)_+.$$

Here the subscript $(\cdot)_+$ denotes the projection onto the nonnegative powers of λ in the Laurent expansion. This motivates the definition of the AKNS hierarchy by prescribing the time evolution

$$(1.18) \quad \Phi_{t_n} = (-i\lambda^n \Psi)_+ \Phi \equiv V_n \Phi.$$

The first few members of the hierarchy are given by

$$\begin{aligned} n = 2 &\Rightarrow \text{nonlinear Schrödinger (NLS) flow,} \\ n = 3 &\Rightarrow \text{complex mKdV flow,} \Rightarrow \text{mKdV flow.} \end{aligned}$$

In terms of the expansion coefficients Ψ_j , the operators V_n take the form

$$\begin{aligned} V_2 &= -i(\lambda^2 \Psi_0 + \lambda \Psi_1 + \Psi_2), \\ V_3 &= -i(\lambda^3 \Psi_0 + \lambda^2 \Psi_1 + \lambda \Psi_2 + \Psi_3). \end{aligned}$$

The zero-curvature equation $U_t - V_x + [U, V] = 0$ with $V_{\text{mKdV}} = 4V_3$ gives, by collecting the constant terms in λ (i.e., at order $O(1)$),

$$Q_t + 4i\Psi_{3,x} + [Q, -4i\Psi_3] = 0.$$

Taking the off-diagonal part gives $Q_t + 4i\Psi_{3,x}^{\text{off}} = 0$, that is,

$$(1.19) \quad Q_t + 4i \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right)_x = Q_t + Q_{xxx} - 2(Q^3)_x = 0.$$

Since

$$Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad Q^2 = -u^2 \mathbb{I},$$

this reduces to the scalar equation

$$u_t + u_{xxx} + 2(u^3)_x = 0,$$

which is precisely the mKdV equation (1.1).

Supplementary computations:

Using the definitions (1.4) and (1.6) we have

$$(1.20) \quad \Psi = \Phi \sigma_3 \Phi^{-1}.$$

Differentiating Ψ with respect to t_n and using (1.18) gives

$$\begin{aligned} \Psi_{t_n} &= \Phi_{t_n} \sigma_3 \Phi^{-1} - \Phi \sigma_3 \Phi^{-1} \Phi_{t_n} \Phi^{-1} \\ &= [\Phi_{t_n} \Phi^{-1}, \Psi] = [V_n, \Psi]. \end{aligned}$$

Thus

$$(1.21) \quad \Psi_{t_n} = [V_n, \Psi].$$

Next we simplify the zero-curvature equation for the pair (U, V_n) . The zero-curvature equation is

$$U_{t_n} - (V_n)_x + [U, V_n] = 0,$$

and since $U = -i\lambda\sigma_3 + Q$ we have $U_{t_n} = Q_{t_n}$. With the normalization

$$V_n = \alpha_n (-i\lambda^n \Psi)_+$$

we obtain

$$Q_{t_n} = (V_n)_x - [U, V_n] = \alpha_n (-i\lambda^n \Psi_x)_+ - [U, \alpha_n (-i\lambda^n \Psi)_+].$$

Pulling out constants and using $\Psi_x = [U, \Psi]$ gives

$$(1.22) \quad Q_{t_n} = -i\alpha_n \left((\lambda^n [U, \Psi])_+ - [U, (\lambda^n \Psi)_+] \right).$$

Write $U = -i\lambda\sigma_3 + Q$ and split the first term in (1.22):

$$\lambda^n [U, \Psi] = -i\lambda^{n+1} [\sigma_3, \Psi] + \lambda^n [Q, \Psi].$$

Because Q does not depend on λ we have

$$(\lambda^n [Q, \Psi])_+ = [Q, (\lambda^n \Psi)_+],$$

so the Q -commutator contributions cancel in (1.22). The remaining terms therefore come only from the σ_3 -commutator:

$$\begin{aligned} Q_{t_n} &= -i\alpha_n \left(-i(\lambda^{n+1} [\sigma_3, \Psi])_+ + i\lambda [\sigma_3, (\lambda^n \Psi)_+] \right) \\ &= \alpha_n \left(\lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ \right). \end{aligned}$$

To simplify the difference of finite sums, use the expansions

$$\Psi = \sum_{j \geq 0} \Psi_j \lambda^{-j}, \quad (\lambda^n \Psi)_+ = \sum_{j=0}^n \lambda^{n-j} \Psi_j.$$

A direct comparison of coefficients shows

$$\begin{aligned} \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ &= \sum_{j=0}^n \lambda^{n+1-j} [\sigma_3, \Psi_j] - \sum_{j=0}^{n+1} \lambda^{n+1-j} [\sigma_3, \Psi_j] \\ &= -\lambda^{n+1-(n+1)} [\sigma_3, \Psi_{n+1}] \\ &= -[\sigma_3, \Psi_{n+1}]. \end{aligned}$$

Hence the evolution of Q is

$$(1.23) \quad Q_{t_n} = -\alpha_n [\sigma_3, \Psi_{n+1}].$$

Since diagonal parts commute with σ_3 and the commutator doubles the off-diagonal part, we may equivalently write

$$(1.24) \quad Q_{t_n} = -2\alpha_n \sigma_3 \Psi_{n+1}^{\text{off}}.$$

Objectives: Construct the L matrix for mKdV, establish its symmetries and conserved determinant, and relate the eigen-data to the associated hyperelliptic curve and Riemann–Theta functions.

Recall the AKNS spectral problem (1974)

$$(2.1) \quad \Phi_x = U(\lambda; Q) \Phi, \quad U(\lambda; Q) = -i\lambda\sigma_3 + Q,$$

where $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and Q is off-diagonal. In what follows we set

$$(2.2) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

Introduce the gauge

$$(2.3) \quad \Phi = m(x; \lambda) e^{-i\lambda x \sigma_3}, \quad m = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + O(\lambda^{-3}),$$

$$(2.4) \quad \Psi := m \sigma_3 m^{-1} = \Psi_0 + \Psi_1 \lambda^{-1} + \Psi_2 \lambda^{-2} + \dots,$$

so that $\Psi^2 = \mathbb{I}$ and Ψ satisfies the stationary zero-curvature equation

$$(2.5) \quad \Psi_x = [U, \Psi].$$

Decompose a matrix A into diagonal and off-diagonal parts $A = A^{\text{diag}} + A^{\text{off}}$. From (2.5) and $\Psi^2 = \mathbb{I}$ one obtains the recursion

$$(2.6) \quad \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3 \left(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}] \right), \quad j \geq 0,$$

$$(2.7) \quad \Psi_{j+1}^{\text{diag}} = -\frac{1}{2}\sigma_3 \sum_{k=1}^j \left(\Psi_k^{\text{diag}} \Psi_{j-k+1}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k+1}^{\text{off}} \right), \quad j \geq 1.$$

Hence each Ψ_j is a differential polynomial in Q and its x -derivatives.

One checks that

$$(2.8) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i(\lambda \Psi)_+.$$

Let $\Phi(x, t_n; \lambda) = m(x, t_n; \lambda) e^{-i(\lambda x + \lambda^n t_n) \sigma_3}$, then

$$(2.9) \quad \Phi_{t_n} \Phi^{-1} = m_{t_n} m^{-1} - i\lambda^n m \sigma_3 m^{-1} = -i(\lambda^n \Psi)_+ \equiv V_n,$$

so that $V_n = -i(\lambda^n \Psi)_+$ and

$$(2.10) \quad \Psi_{t_n} = [V_n, \Psi].$$

The compatibility $\Phi_{xt_n} = \Phi_{t_n x}$ is equivalent to the zero curvature equation

$$(2.11) \quad U_{t_n} - V_{n,x} + [U, V_n] = 0.$$

Definition 1. Let $g \geq 0$ be a fixed integer, and let $\{\alpha_j\}_{j=0}^{g+1}$ be constants. Define

$$(2.12) \quad L := -i \sum_{j=0}^{g+1} \alpha_j (\lambda^j \Psi(x, t, \lambda))_+,$$

where $(\cdot)_+$ denotes the projection onto nonnegative powers of λ .

It is convenient to rewrite L by expanding the nonnegative part explicitly. Using $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ we have

$$(\lambda^j \Psi)_+ = \sum_{k=0}^j \Psi_k \lambda^{j-k},$$

hence

$$(2.13) \quad L(\lambda) = -i \sum_{j=0}^{g+1} \alpha_j \sum_{k=0}^j \Psi_k \lambda^{j-k} = -i \sum_{l=0}^{g+1} \left(\sum_{j=l}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$ to obtain the last form.

Proposition 1. *Suppose L as defined in (2.12) satisfies the stationary zero-curvature equations*

$$(2.14) \quad L_x = [U, L], \quad L_{t_n} = [V_n, L].$$

Then the following additional constraint holds:

$$(2.15) \quad \sum_{j=1}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}}(x, t_n) + i \alpha_0 Q = 0.$$

Proof. Set

$$(2.16) \quad \widehat{L}(\lambda) := -i \left(\sum_{j=0}^{g+1} \alpha_j \lambda^j \right) \Psi(\lambda).$$

By construction $\widehat{L}_x = [U, \widehat{L}]$. Expand $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ and substitute into $\widehat{L}$:

$$(2.17) \quad \widehat{L}(\lambda) = -i \sum_{j=0}^{g+1} \sum_{k \geq 0} \alpha_j \Psi_k \lambda^{j-k} = -i \sum_{l=-\infty}^{g+1} \left(\sum_{j=0}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$. Decompose $\widehat{L} = \widehat{L}_+ + \widehat{L}_-$ into nonnegative and negative powers of λ . From $\widehat{L}_x = [U, \widehat{L}]$ we obtain

$$\left(\widehat{L}_+ + \widehat{L}_- \right)_x = [U, \widehat{L}_+ + \widehat{L}_-].$$

After extracting nonnegative powers of λ , we have

$$(\widehat{L}_+)_x - [U, \widehat{L}_+] = \left([-i\lambda\sigma_3, \widehat{L}_-] \right)_+ = -i \left[\sigma_3, \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1} \right] = -2i\sigma_3 \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}} = 0,$$

which yields the constraint (2.15).

Next, to ensure compatibility, we require to prove $L_{xt_n} = L_{t_n x}$. Observe that

$$\begin{aligned} L_{xt_n} &= [U_{t_n}, L] + [U, [V_n, L]], \\ L_{t_n x} &= [V_{n,x}, L] + [V_n, [U, L]]. \end{aligned}$$

Applying the Jacobi identity,

$$[[A, B], C] + [[C, A], B] + [[B, C], A] = 0,$$

we find

$$[U, [V_n, L]] - [V_n, [U, L]] = [[L, V_n], U] + [[U, L], V_n] = [[U, V_n], L].$$

Therefore, it follows from the zero-curvature condition (2.11) that

$$L_{xt_n} - L_{t_n x} = [U_{t_n} - V_{n,x} + [U, V_n], L] = 0.$$

□

Specializing to genus $g = 2$, define

$$(2.18) \quad L(x, t; \lambda) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

$$(2.19) \quad V(\lambda; Q) = 4\lambda^2 U(\lambda; Q) + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

The zero curvature condition for the mKdV equation is

$$U_t - V_x + [U, V] = 0,$$

which yields the scalar modified KdV equation

$$(2.20) \quad u_t + u_{xxx} + 6u^2 u_x = 0, \quad u(x, t) \in \mathbb{R}.$$

The matrices U and V satisfy the following symmetries:

$$(2.21) \quad \begin{aligned} U^\dagger(\lambda^*; Q) &= -U(\lambda; Q), & U^\top(-\lambda; Q) &= -U(\lambda; Q), \\ V^\dagger(\lambda^*; Q) &= -V(\lambda; Q), & V^\top(-\lambda; Q) &= -V(\lambda; Q). \end{aligned}$$

As a consequence the matrix L inherits analogous symmetries:

$$(2.22) \quad L^\dagger(x, t; \lambda^*) = -L(x, t; \lambda), \quad L^\top(x, t; -\lambda) = -L(x, t; \lambda).$$

Proposition 2. *The determinant of L is independent of x and t :*

$$\partial_x \det L = \partial_t \det L = 0.$$

Proof. By Jacobi's formula (or Abel's theorem applying to $L_x = (L_x L^{-1})L$), for any differentiable matrix $L(x, t; \lambda)$

$$\partial_x \det L = \det L \operatorname{tr}(L_x L^{-1}).$$

Using the stationary zero-curvature equation $L_x = [U, L] = UL - LU$ we obtain

$$\partial_x \det L = \det L \operatorname{tr}((UL - LU)L^{-1}) = \det L (\operatorname{tr} U - \operatorname{tr}(L^{-1}LU)) = 0,$$

since $\operatorname{tr}(LUL^{-1}) = \operatorname{tr}(L^{-1}LU)$. The identical argument with $\partial_t L = [V, L]$ yields $\partial_t \det L = 0$. Hence $\det L$ does not depend on x or t . $\square$

Therefore the determinant of L is constant in x and t . Write its factorization as

$$(2.23) \quad \det L(\lambda) = \prod_{j=1}^6 (\lambda - \hat{\lambda}_j).$$

By the symmetry relations (2.22) each root $\hat{\lambda}_j$ forces $\hat{\lambda}_j^*$ and $-\hat{\lambda}_j$ to be roots as well. Two possible root structures follow.

Case 1. Three purely imaginary pairs:

$$(2.24) \quad \det L(\lambda) = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2), \quad \lambda_j \in i\mathbb{R}.$$

Case 2. One nonreal quadruple together with one purely imaginary pair:

$$(2.25) \quad \det L(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_1^*)(\lambda + \lambda_1)(\lambda + \lambda_1^*)(\lambda - \lambda_2)(\lambda + \lambda_2),$$

with $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$ and $\lambda_2 \in i\mathbb{R}$.

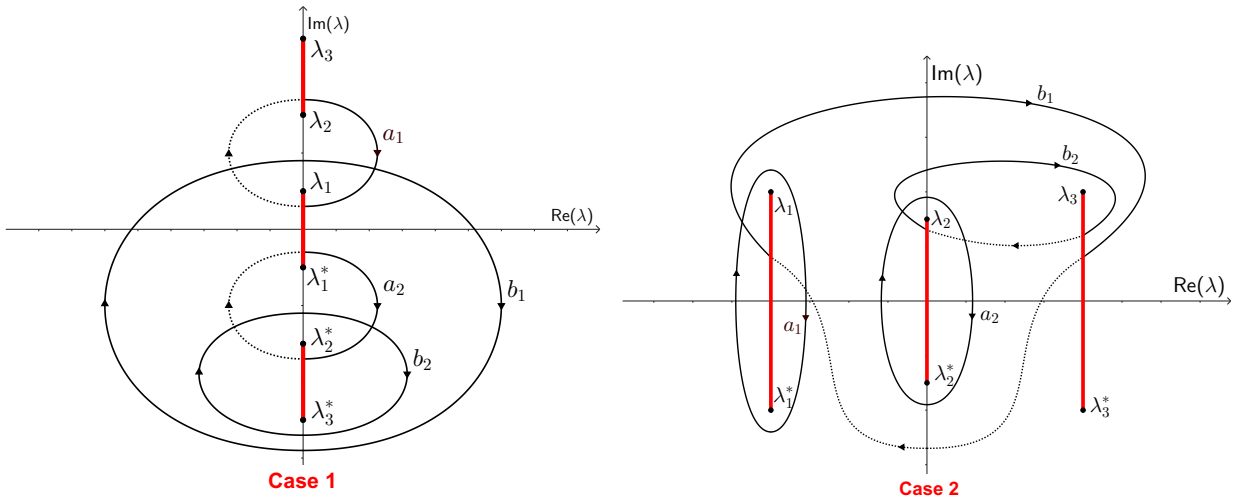


FIGURE 1. The homology basis for the curve $y^2 = \prod_{i=1}^3 (\lambda^2 - \lambda_i^2)$.

The matrix elements of L take the explicit form

$$\begin{aligned}
(2.26) \quad L_{11} &= -i\lambda^3 - i\lambda\left(\alpha_1 - \frac{1}{2}u^2\right), \quad L_{22} = -L_{11}, \\
L_{12} &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\
L_{21} &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*),
\end{aligned}$$

where μ_1, μ_2 denote the (in general complex) roots of the quadratic factor in L_{12}/u . The conjugation in L_{21} is consistent with the reality of u and the symmetry relations (2.22).

Recall that

$$\begin{cases} \Phi_x = U\Phi, \\ \Phi_t = V\Phi, \end{cases} \quad \begin{cases} L_x = [U, L], \\ L_t = [V, L], \end{cases}$$

It follows that $L\Phi$ also satisfies the same Lax pair, since

$$\begin{aligned}
(L\Phi)_x &= L_x\Phi + L\Phi_x = (UL - LU)\Phi + LU\Phi = U(L\Phi), \\
(L\Phi)_t &= V(L\Phi).
\end{aligned}$$

This observation is also a direct consequence of the identity $L = \Psi = \Phi\sigma_3\Phi^{-1}$ which implies $L\Phi = \Phi\sigma_3$.

Assume $\Phi(x, t; \lambda)$ is normalized by $\Phi(0, 0; \lambda) = \mathbb{I}$. By uniqueness of solutions of the linear ODEs we obtain

$$(2.27) \quad L(x, t; \lambda) \Phi(x, t; \lambda) = \Phi(x, t; \lambda) L(0, 0; \lambda).$$

In particular, evaluation at $(x, t) = (0, 0)$ yields $L(0, 0; \lambda)$. So knowledge of $L(0, 0; \lambda)$ determines the action of L on the columns of Φ for all (x, t) .

Set

$$(2.28) \quad \det(L(x, t; \lambda)) = \det(L(0, 0; \lambda)) = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = y^2, \quad y = \pm \sqrt{\prod_{j=1}^3 (\lambda^2 - \lambda_j^2)}.$$

A short calculation shows that $\pm iy$ are eigenvalues of $L(0, 0; \lambda)$:

$$\det(\pm iy \mathbb{I} - L(0, 0; \lambda)) = -y^2 - L_{11}^2 - L_{12}L_{21} = -y^2 + \det(L(0, 0; \lambda)) = 0,$$

where L_{ij} denotes the (i, j) -entry of $L(0, 0; \lambda)$.

Hence there exists an invertible matrix M , independent of x and t , such that

$$(2.29) \quad L(0, 0; \lambda) = M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix} M^{-1}.$$

Combining (2.27) and (2.29) yields

$$L(x, t; \lambda) \Phi(x, t; \lambda) M = \Phi(x, t; \lambda) M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix}.$$

Write the two columns of $\Phi(x, t; \lambda)M$ as

$$\Phi(x, t; \lambda)M = (\Phi_1, \Phi_2),$$

so that

$$L(x, t; \lambda)\Phi_1 = iy\Phi_1, \quad L(x, t; \lambda)\Phi_2 = -iy\Phi_2.$$

Equivalently,

$$(L - iy)\Phi_1 = 0, \quad (L + iy)\Phi_2 = 0,$$

and therefore

$$\ker(L \mp iy) = \text{span}\{\Phi_{1,2}\}.$$

Since each eigenvector may be rescaled, we may write

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \Phi_{11} \begin{pmatrix} 1 \\ r_1 \end{pmatrix}, \quad \Phi_2 = \Phi_{12} \begin{pmatrix} 1 \\ r_2 \end{pmatrix},$$

so

$$\ker(L \mp iy) = \text{span} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix}.$$

Writing the eigenvalue equation explicitly gives

$$\begin{pmatrix} L_{11} \mp iy & L_{12} \\ L_{21} & -L_{11} \mp iy \end{pmatrix} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix} = 0,$$

from which one obtains the two equivalent expressions for the ratio $r_{1,2}$:

$$(2.30) \quad r_{1,2} = \frac{\pm iy - L_{11}}{L_{12}} = \frac{L_{21}}{L_{11} \pm iy}.$$

The point $P = (\lambda, y)$ lies on the hyperelliptic curve (2.28), which is a compact Riemann surface of genus $g = 2$ with punctures $\infty^\pm$. Here $P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3$. Denote

$$\begin{aligned} \mu &= \{\hat{\mu}_1, \hat{\mu}_2\}, & \hat{\mu}_j &= (\mu_j, iL_{11}(\mu_j)), & j &= 1, 2, \\ \mu^* &= \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, & \hat{\mu}_j^* &= (\mu_j^*, -iL_{11}(\mu_j^*)), & j &= 1, 2. \end{aligned}$$

The divisor of r_1 is then given by

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

That is, $r_1(x, t; P)$ has zeros at the points of μ^* and ∞^- , and poles at the points of μ and ∞^+ . This description of the divisor is central in constructing Baker–Akhiezer functions on the hyperelliptic Riemann surface defined by

$$y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2).$$

Definition 2 (Riemann–Theta function). *Let R_g be a compact Riemann surface of genus g . Fix a canonical homology basis and let B denote the associated Riemann period matrix, which is symmetric with negative definite real part. The Riemann–Theta function is the entire function*

$$\Theta(B; z) = \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top z \right\},$$

where

$$n = \begin{pmatrix} n_1 \\ \vdots \\ n_g \end{pmatrix}, \quad n_j \in \mathbb{Z}, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_g \end{pmatrix}, \quad z_j \in \mathbb{C}.$$

Proposition 3. *The Riemann–Theta function satisfies:*

(1) *Periodicity in the standard lattice directions:*

$$\Theta(z + 2\pi i e_k) = \Theta(z), \quad k = 1, \dots, g.$$

(2) *Quasi-periodicity in the period matrix directions:*

$$\Theta(z + B e_k) = \Theta(z) e^{-z_k - B_{kk}/2}, \quad k = 1, \dots, g.$$

(3) *Evenness:*

$$\Theta(-z) = \Theta(z).$$

Proof. (1) For $z \mapsto z + 2\pi i e_k$, the exponential in the definition acquires a factor

$$\exp(n^\top (2\pi i e_k)) = \exp(2\pi i n_k) = 1,$$

since $n_k \in \mathbb{Z}$. Thus Θ is periodic in this direction.

(2) For $z \mapsto z + B e_k$, we compute

$$\begin{aligned}\Theta(z + B e_k) &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top (z + B e_k)\right\} \\ &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top z + n^\top B e_k\right\}.\end{aligned}$$

Write $m = n + e_k$. Then $n = m - e_k$, and the exponent becomes

$$\begin{aligned}&\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k \\ &= \frac{1}{2}(m^\top B m - m^\top B e_k - e_k^\top B m + e_k^\top B e_k) + m^\top z - e_k^\top z + m^\top B e_k - e_k^\top B e_k \\ &= \frac{1}{2}m^\top B m + m^\top z - \frac{1}{2}(m^\top B e_k + e_k^\top B m) + m^\top B e_k - \frac{1}{2}e_k^\top B e_k - e_k^\top z.\end{aligned}$$

Since $B = B^\top$, we have $m^\top B e_k = e_k^\top B m$; moreover, $e_k^\top z = z_k$ and $e_k^\top B e_k = B_{kk}$. Hence

$$\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k = \frac{1}{2}m^\top B m + m^\top z - z_k - \frac{1}{2}B_{kk}.$$

Therefore,

$$\Theta(z + B e_k) = e^{-z_k - B_{kk}/2} \Theta(z).$$

(3) Finally, since the defining series is invariant under $n \mapsto -n$, it follows that

$$\Theta(-z) = \Theta(z).$$

□

Supplementary computations:

Since

$$U(\lambda; Q) = -i\lambda\sigma_3 + Q = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix},$$

we have

$$Q^2 = -u^2 \mathbb{I}, \quad Q^3 = -u^2 Q.$$

Compute each term in

$$V(\lambda; Q) = 4\lambda^2 U + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

First,

$$4\lambda^2 U = \begin{pmatrix} -4i\lambda^3 & 4\lambda^2 u \\ -4\lambda^2 u & 4i\lambda^3 \end{pmatrix}, \quad Q_x - Q^2 = \begin{pmatrix} u^2 & u_x \\ -u_x & u^2 \end{pmatrix}.$$

Hence

$$2i\lambda\sigma_3(Q_x - Q^2) = \begin{pmatrix} 2i\lambda u^2 & 2i\lambda u_x \\ 2i\lambda u_x & -2i\lambda u^2 \end{pmatrix}.$$

Since $Q_{xx} - 2Q^3 = Q_{xx} + 2u^2 Q$,

$$-(Q_{xx} - 2Q^3) = \begin{pmatrix} 0 & -(u_{xx} + 2u^3) \\ u_{xx} + 2u^3 & 0 \end{pmatrix}.$$

Adding all contributions gives

$$V(\lambda; Q) = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix},$$

where

$$(2.31) \quad V_{11} = -4i\lambda^3 + 2i\lambda u^2, \quad V_{22} = -V_{11},$$

$$(2.32) \quad V_{12} = 4\lambda^2 u + 2i\lambda u_x - (u_{xx} + 2u^3),$$

$$(2.33) \quad V_{21} = -4\lambda^2 u + 2i\lambda u_x + (u_{xx} + 2u^3).$$

Week 3: Algebro-Geometric Solutions for the mKdV Equation (3)

2025-09-28

Objectives: Construct algebro-geometric mKdV solutions on a genus-two curve by expressing r_1 and u in Riemann theta functions and matching their asymptotics at $\infty^\pm$, poles and zeroes.

The Riemann–Roch theorem provides a fundamental relation between the divisor of a meromorphic function and the corresponding Abel map data on a Riemann surface. In the present setting, for the meromorphic function $r_1(x, t; P)$ we have

$$(3.1) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda),$$

which identifies the divisor of r_1 as the difference between two effective divisors supported at the points $\infty^\pm$ and at the divisor of spectral data μ, μ^* .

A concrete representation of the function r_1 is given in terms of Riemann theta functions:

$$(3.2) \quad r_1(x, t; P) = \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu}))} \exp(\Omega_1(P) - \ln(\omega_1)), \quad P = (\lambda, y).$$

Here Θ denotes the Riemann theta function, $\mathcal{C}$ is the vector of Riemann constants, $\mathcal{A}_{P_0}$ is the Abel map based at P_0 , and α_{P_0} is its linear extension to divisors. The exponential factor is determined by an Abelian integral Ω_1 , while $\hat{r}_1(x, t)$ is a normalizing prefactor independent of λ .

The underlying algebraic curve R_2 associated with this construction is the genus-two hyperelliptic Riemann surface

$$(3.3) \quad R_2 := \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3 + O(\lambda).$$

Thus each point $P \in R_2$ is represented by (λ, y) , where

$$y = \pm \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

To construct the Jacobi variety one chooses a canonical homology basis $\{a_1, a_2; b_1, b_2\}$ on R_2 , satisfying

$$(3.4) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = -b_j \circ a_i = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 1, 2.$$

A basis of holomorphic differentials is given by

$$(3.5) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}, \quad d_{ik} \in \mathbb{C},$$

where the normalization guarantees the a -periods form the identity matrix and the b -periods form the Riemann period matrix $B = (B_{ij})$.

The Abel map is defined using these normalized differentials. First, introduce the period lattice

$$(3.6) \quad L_2 = \{z \in \mathbb{C}^2 \mid z = 2\pi i(n + Bm), \quad n, m \in \mathbb{Z}^2\},$$

where B is the period matrix and n, m, z are column vectors in $\mathbb{C}^2$. The Jacobian of R_2 is then

$$(3.7) \quad J(R_2) = \mathbb{C}^2 / L_2.$$

For a base point $P_0 \in R_2$, the Abel map sends a point $P \in R_2$ to

$$(3.8) \quad \mathcal{A}_{P_0} : R_2 \rightarrow J(R_2),$$

$$(3.9) \quad P \mapsto \mathcal{A}_{P_0}(P) = (\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P)) = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

This definition extends linearly to divisors. A divisor $D = \sum_{P \in R_2} D(P) \cdot P$ is a finite formal sum of points P on R_2 with integer coefficients $D(P) \in \mathbb{Z}$, and we set

$$(3.10) \quad \alpha_{P_0}(D) = \sum_{P \in R_2} D(P) \mathcal{A}_{P_0}(P).$$

Finally, we introduce the Abelian integral

$$(3.11) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1, \quad \oint_{a_i} d\Omega_1 = 0,$$

with the prescribed asymptotics at the points at infinity,

$$(3.12) \quad \Omega_1(P) = \pm \left(\ln(\lambda) + \ln(\omega_1) + o(1) \right), \quad P \rightarrow \infty^\pm.$$

This integral provides the exponential term appearing in (3.2) and ensures the correct divisor structure of r_1 .

Recall that the function $r_1(x, t; P)$ admits the representation

$$(3.13) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}},$$

where

$$(3.14) \quad L_{21} = -u \left(\lambda^2 - \frac{i u_x}{2u} \lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2} \right), \quad L_{11} = -i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right).$$

To analyze its asymptotic behavior, note first that on the hyperelliptic curve R_2 ,

$$(3.15) \quad y = \pm \left(\lambda^3 - \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda \right) + O(\lambda^{-1}), \quad P \rightarrow \infty^\pm.$$

As $P \rightarrow \infty^-$, the expansion of the denominator yields

$$(3.16) \quad L_{11} + iy = -2i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right) + \frac{i}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda + O(\lambda^{-1}),$$

so that

$$(3.17) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}} = -\frac{i u}{2\lambda} + O(\lambda^{-2}), \quad P \rightarrow \infty^-.$$

Similarly, near ∞^+ one obtains

$$(3.18) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u}\lambda + O(1), \quad P \rightarrow \infty^+.$$

On the other hand, from the Riemann theta function representation (3.2), the asymptotics at the two points at infinity are

$$(3.19) \quad r_1(x, t; P) = \begin{cases} \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))} \lambda + O(1), & P \rightarrow \infty^+, \\ \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} \lambda^{-1} + O(\lambda^{-2}), & P \rightarrow \infty^-. \end{cases}$$

By comparing these expansions with the direct computation, one obtains the following theta-function formula for the potential $u(x, t)$:

$$(3.20) \quad u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

This identity expresses the potential u in terms of theta functions associated with the Jacobian of the spectral curve, thereby linking the algebro-geometric spectral data with the physical field variable.

Recall that the column vector

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}$$

satisfies both parts of the Lax pair:

$$(3.21) \quad \partial_x \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}, \quad \partial_t \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & -V_{11} \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}.$$

Hence the scalar component Φ_{11} satisfies

$$(3.22) \quad \Phi_{11,x} = (-i\lambda + u r_1) \Phi_{11}, \quad \Phi_{11,t} = (V_{11} + V_{12} r_1) \Phi_{11}.$$

Integrating the differential 1-form in (3.22) from the base point (x_0, t_0) to (x, t) yields

$$(3.23) \quad \Phi_{11}(x, t; \lambda) = \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{(x_0, t_0)}^{(x, t)} \left[(-i\lambda + u r_1) dx + (V_{11} + V_{12} r_1) dt \right] \right).$$

Hence $\Phi_{11} \in \mathcal{M}(R_2 \setminus \{\infty^\pm\})$, the space of functions meromorphic on R_2 away from $\infty^\pm$. The representation (3.23) also shows that Φ_{11} has an essential singularity at each of $\infty^\pm$. For convenience, we set the initial point as $(x_0, t_0) = (0, 0)$.

The large- λ asymptotics differ on the two sheets. On the sheet with $P \rightarrow \infty^-$ we fix the normalization

$$\Phi_{1,0}^- := \lim_{P \rightarrow \infty^-} \Phi_{11}(x_0, t_0; \lambda) = 1,$$

and obtain

$$(3.24) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \exp(-i\lambda(x + 4\lambda^2 t)) + O(1), & P \rightarrow \infty^-, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = O(\lambda^{-1}) \exp(-i\lambda(x + 4\lambda^2 t)), & P \rightarrow \infty^-. \end{aligned}$$

On the other sheet, $P \rightarrow \infty^+$, the expansion of r_1 is

$$(3.25) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u} \lambda + \frac{u_x}{u^2} + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Consequently

$$(3.26) \quad -i\lambda + u r_1 = i\lambda + (\ln u)_x + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Substituting (3.26) into (3.23) gives, with the normalization constant

$$(3.27) \quad \Phi_{1,0}^+ := \lim_{P \rightarrow \infty^+} \Phi_{11}(x_0, t_0; \lambda),$$

the asymptotic formulas

$$(3.28) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{1,0}^+ u(x, t) \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = 2i\lambda \Phi_{1,0}^+ \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+. \end{aligned}$$

Recall

$$(3.29) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)},$$

$$(3.30) \quad r_1(\lambda) = \frac{L_{21}(\lambda)}{iy(\lambda) + L_{11}(\lambda)} = \frac{-u(\lambda - \mu_1^*)(\lambda - \mu_2^*)}{iy + L_{11}}.$$

We write

$$L(\lambda) = \begin{pmatrix} L_{11}(\lambda) & L_{12}(\lambda) \\ L_{21}(\lambda) & -L_{11}(\lambda) \end{pmatrix}, \quad U(\lambda) = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix},$$

and recall the stationary zero-curvature equation

$$(3.31) \quad L_x = [U, L].$$

With $\sigma_3 = \text{diag}(1, -1)$ and off-diagonal part $L^{\text{off}} = \begin{pmatrix} 0 & L_{12} \\ L_{21} & 0 \end{pmatrix}$, we compute

$$\begin{aligned} [U, L] &= -2i\lambda \sigma_3 L^{\text{off}} + [Q, L] \\ &= -2i\lambda \sigma_3 L^{\text{off}} + \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} - \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \\ &= -2i\lambda \begin{pmatrix} 0 & L_{12} \\ -L_{21} & 0 \end{pmatrix} + \begin{pmatrix} u(L_{21} + L_{12}) & -2uL_{11} \\ -2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix} \\ (3.32) \quad &= \begin{pmatrix} u(L_{21} + L_{12}) & -2i\lambda L_{12} - 2uL_{11} \\ 2i\lambda L_{21} - 2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix}. \end{aligned}$$

Thus from (3.31) we obtain

$$\begin{aligned} (3.33) \quad \partial_x L_{11} &= u(L_{21} + L_{12}), \\ \partial_x L_{12} &= -2i\lambda L_{12} - 2uL_{11}, \\ \partial_x L_{21} &= 2i\lambda L_{21} - 2uL_{11}. \end{aligned}$$

By assumption

$$(3.34) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2).$$

Differentiating with respect to x gives

$$(3.35) \quad \partial_x L_{12}(\lambda) = u_x(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,x}(\lambda - \mu_2) - u \mu_{2,x}(\lambda - \mu_1).$$

Evaluating (3.35) at $\lambda = \mu_1$ leaves only the middle term:

$$(3.36) \quad \partial_x L_{12}(\mu_1) = -u \mu_{1,x}(\mu_1 - \mu_2).$$

On the other hand, from (3.33) evaluated at $\lambda = \mu_1$ we have

$$(3.37) \quad \partial_x L_{12}(\mu_1) = -2u L_{11}(\mu_1).$$

Equating the two expressions yields

$$(3.38) \quad \mu_{1,x} = \frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2}.$$

Next, the determinant relation $y^2 = \det L = -L_{11}^2 - L_{12}L_{21}$ at $\lambda = \mu_1$ reduces to

$$(3.39) \quad y(\mu_1)^2 = -L_{11}(\mu_1)^2,$$

since $L_{12}(\mu_1) = 0$. Hence $-iy(\mu_1) = \pm L_{11}(\mu_1)$. We fix the sign by setting

$$(3.40) \quad -iy(\mu_1) = L_{11}(\mu_1),$$

so that $P = (\mu_1, iL_{11}(\mu_1))$ represents a pole of the meromorphic function $r_1(P) = r_1(\lambda)$ on the Riemann surface R_2 .

Consider now the principal part of $u r_1$ near $\lambda = \mu_1$. From (3.29) and (3.34),

$$(3.41) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_1) - L_{11}(\mu_1)}{(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using (3.40) and (3.38),

$$iy(\mu_1) - L_{11}(\mu_1) = -2L_{11}(\mu_1) = -(\mu_1 - \mu_2) \mu_{1,x}.$$

Thus (3.41) becomes

$$(3.42) \quad u r_1(\lambda) = -\frac{\mu_{1,x}}{\lambda - \mu_1} + O(1) = \partial_x(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Then by (3.23) and (3.42) we conclude that $\lambda = \mu_1$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

Supplementary computations:

Evaluate (3.35) at $\lambda = \mu_2$. Only the third term survives, so

$$(3.43) \quad \partial_x L_{12}(\mu_2) = -u \mu_{2,x}(\mu_2 - \mu_1).$$

From (3.33) at $\lambda = \mu_2$ we have

$$(3.44) \quad \partial_x L_{12}(\mu_2) = -2u L_{11}(\mu_2).$$

Equating these expressions yields

$$(3.45) \quad \mu_{2,x} = \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1}.$$

Adopting the same sheet convention as for μ_1 we take

$$(3.46) \quad -iy(\mu_2) = L_{11}(\mu_2).$$

Compute the principal part of $u r_1$ near $\lambda = \mu_2$. From (3.29) and (3.34),

$$(3.47) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_2) - L_{11}(\mu_2)}{(\mu_2 - \mu_1)} \cdot \frac{1}{\lambda - \mu_2} + O(1), \quad \lambda \rightarrow \mu_2.$$

Using (3.46) and (3.45) we obtain

$$iy(\mu_2) - L_{11}(\mu_2) = -2L_{11}(\mu_2) = -(\mu_2 - \mu_1) \mu_{2,x}.$$

Hence (3.47) becomes

$$(3.48) \quad u r_1(\lambda) = -\frac{\mu_{2,x}}{\lambda - \mu_2} + O(1) = \partial_x(\ln(\lambda - \mu_2)) + O(1), \quad \lambda \rightarrow \mu_2.$$

Therefore, by (3.23) and (3.48), $\lambda = \mu_2$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

We next consider the time evolution governed by

$$(3.49) \quad L_t = [V, L], \quad V(\lambda) = \begin{pmatrix} V_{11}(\lambda) & V_{12}(\lambda) \\ V_{21}(\lambda) & -V_{11}(\lambda) \end{pmatrix}.$$

Direct computation of the commutator yields

$$[V, L] = VL - LV = \begin{pmatrix} V_{12}L_{21} - L_{12}V_{21} & 2V_{11}L_{12} - 2L_{11}V_{12} \\ 2L_{11}V_{21} - 2V_{11}L_{21} & L_{12}V_{21} - V_{12}L_{21} \end{pmatrix},$$

since the trace vanishes and the anti-diagonal structure pairs terms. Consequently from (3.49) we obtain the componentwise equations

$$(3.50) \quad \begin{aligned} \partial_t L_{11} &= V_{12}L_{21} - L_{12}V_{21}, \\ \partial_t L_{12} &= 2V_{11}L_{12} - 2L_{11}V_{12}, \\ \partial_t L_{21} &= 2L_{11}V_{21} - 2V_{11}L_{21}. \end{aligned}$$

By assumption

$$(3.51) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2),$$

where $u = u(x, t)$ and $\mu_j = \mu_j(x, t)$. Differentiating (3.51) with respect to t gives

$$(3.52) \quad \partial_t L_{12}(\lambda) = u_t(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,t}(\lambda - \mu_2) - u \mu_{2,t}(\lambda - \mu_1).$$

Evaluating (3.52) at $\lambda = \mu_1$ annihilates the first and third terms and leaves

$$(3.53) \quad \partial_t L_{12}(\mu_1) = -u \mu_{1,t}(\mu_1 - \mu_2).$$

On the other hand, evaluating the second equation in (3.50) at $\lambda = \mu_1$ yields

$$(3.54) \quad \partial_t L_{12}(\mu_1) = 2V_{11}(\mu_1) L_{12}(\mu_1) - 2L_{11}(\mu_1) V_{12}(\mu_1).$$

Since $L_{12}(\mu_1) = 0$ by (3.51), the first term on the right of (3.54) vanishes and we obtain

$$(3.55) \quad \partial_t L_{12}(\mu_1) = -2L_{11}(\mu_1) V_{12}(\mu_1).$$

Equating (3.53) and (3.55) gives the evolution law for μ_1 :

$$(3.56) \quad \mu_{1,t} = \frac{2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)}.$$

Consider the principal part of $V_{12}(\lambda) r_1(\lambda)$ near $P = (\mu_1, iL_{11}(\mu_1))$. From (3.29) we have

$$(3.57) \quad V_{12}(\lambda) r_1(\lambda) = V_{12}(\lambda) \frac{iy(\lambda) - L_{11}(\lambda)}{u(\lambda - \mu_1)(\lambda - \mu_2)}.$$

Expanding the numerator and evaluating at $\lambda = \mu_1$ yields the simple pole

$$(3.58) \quad V_{12}(\lambda) r_1(\lambda) = \frac{-2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using the evolution law (3.56) for μ_1 this principal part becomes

$$(3.59) \quad V_{12}(\lambda) r_1(\lambda) = -\frac{\mu_{1,t}}{\lambda - \mu_1} + O(1) = \partial_t(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Week 4: Algebraic-Geometric Solutions for the mKdV Equation (4)

2025-10-14

Objectives: Continue to construct algebraic-geometric mKdV solutions on a genus-two curve by expressing r_1 and u in Riemann theta functions and matching their asymptotics at $\infty^\pm$, poles and zeroes.

Recall $P = (\lambda, y)$. The genus-two hyperelliptic curve and its points at infinity are

$$(4.1) \quad R_2 = \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \iff \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3.$$

Define the points

$$(4.2) \quad \mu = \{\hat{\mu}_1, \hat{\mu}_2\}, \quad \hat{\mu}_j = (\mu_j, iL_{11}(\mu_j)), \quad \mu^* = \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, \quad \hat{\mu}_j^* = (\mu_j^*, -iL_{11}(\mu_j^*)),$$

so that μ^* are the zeros and μ the poles of r_1 .

At a pole $\hat{\mu}_1 = (\mu_1, iL_{11}(\mu_1))$ the meromorphic function $r_1(\lambda)$ has the local expansion

$$(4.3) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{-2L_{11}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Recall that $r_1(x, t; P)$ is defined in (3.2) and that $\Phi_{11}(x, t; P) = \Phi_{11}(x, t; \lambda)$ is given by the integral representation (3.23). Choosing a continuous path from (x_0, t_0) to (x, t) we have

$$(4.4) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{(x_0, t_0)}^{(x, t)} (-i\lambda + u r_1) dx_1 + (V_{11} + V_{12} r_1) dt_1 \right) \\ &= \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{x_0}^x (-i\lambda + u r_1)(x_1, t_0) dx_1 + \int_{t_0}^t (V_{11} + V_{12} r_1)(x, t_1) dt_1 \right). \end{aligned}$$

The singularity structure of Φ_{11} follows from the pole structure of r_1 . From (4.3) and the explicit expressions for $u r_1$ and $V_{12} r_1$ (see (3.42) and (3.48)) one obtains, as $\lambda \rightarrow \mu_j$,

$$(4.5) \quad u r_1(x_1, t_0; \lambda) = \partial_{x_1}(\ln(\lambda - \mu_j(x_1, t_0))) + O(1),$$

and

$$(4.6) \quad V_{12} r_1(x, t_1; \lambda) = \partial_{t_1}(\ln(\lambda - \mu_j(x, t_1))) + O(1).$$

Therefore each moving point $\hat{\mu}_j(x, t) = (\mu_j(x, t), iL_{11}(\mu_j(x, t)))$ is a zero of $\Phi_{11}(x, t; P)$, while the initial points $\mu_0 := \mu(x_0, t_0)$ give the poles of Φ_{11} .

Matching the asymptotic forms (3.24) and (3.28) of Φ_{11} as $P \rightarrow \infty^\pm$ gives the theta-function representation

$$(4.7) \quad \Phi_{11}(x, t; P) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{i(\omega_2 + \Omega_2(P))x + i(\omega_3 + \Omega_3(P))t}.$$

Here $P_0 = (\lambda_3, 0)$, and the Abelian integrals Ω_2, Ω_3 are defined by

$$(4.8) \quad \Omega_2(P) := \int_{P_0}^P d\Omega_2, \quad \oint_{a_i} d\Omega_2 = 0, \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad P \rightarrow \infty^\pm,$$

$$(4.9) \quad \Omega_3(P) := \int_{P_0}^P d\Omega_3, \quad \oint_{a_i} d\Omega_3 = 0, \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

where ω_2, ω_3 are constants independent of λ . As $P \rightarrow \infty^-$, (4.7) is consistent with (3.24).

As $P \rightarrow \infty^+$, comparing (3.28) with (4.7) yields

$$(4.10) \quad u(x, t) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Combining (3.20) and (4.10), we start from the first expression in (3.20):

$$u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}.$$

Equating this with (4.10) gives

$$2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Cancel the common factor $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))$ and solve for $\hat{r}_1(x, t)$:

$$(4.11) \quad \hat{r}_1(x, t) = \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

Substitute (4.11) into the second form of (3.20),

$$u(x, t) = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

Then

$$\begin{aligned} u(x, t) &= 2i \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t} \\ &\quad \times \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}. \end{aligned}$$

After cancelling $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))$, we obtain

$$(4.12) \quad u(x, t) = \frac{(2i)^2 \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

We need to determine these the parameters:

$$\alpha_{P_0}(D_\mu), \quad \alpha_{P_0}(D_{\mu^*}), \quad B, \quad w_1, w_2, \quad \omega_1, \omega_2, \omega_3, \quad \Phi_{1,0}^+.$$

The Riemann-Roch theorem gives

$$(4.13) \quad r(-D) = \deg(D) + i(D) - g + 1,$$

where

$$\mathcal{L}(D) = \{f \in \mathcal{M}(R_g) \mid (f) \geq D\}, \quad r(D) = \dim(\mathcal{L}(D)),$$

and

$$\mathcal{L}'(D) = \{\omega \in \mathcal{M}'(R_g) \mid (\omega) \geq D\}, \quad i(D) = \dim(\mathcal{L}'(D)),$$

with $\mathcal{M}(R_g)$ the field of meromorphic functions on the genus g curve R_g and $\mathcal{M}'(R_g)$ the space of meromorphic differentials.

Recall that

$$(4.14) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

Take the (non special) divisor $D = D_{\infty^+, \mu}$. Assume $i(D) = 0$ and that $(r_1) \geq -D_{\infty^+, \mu}$. Then by (4.13) (with $g = 2$ and $\deg(D) = 3$ in the present setup)

$$(4.15) \quad r(-D_{\infty^+, \mu}) = \deg(D) - g + 1 = 3 - 2 + 1 = 2.$$

Since r_1 and $\tilde{r}_1$ have the same zeros they span a two dimensional space and hence r_1 is proportional to $\tilde{r}_1$, i.e.

$$r_1 = c\tilde{r}_1 + d \Rightarrow r_1 = c\tilde{r}_1 \quad (c, d \in \mathbb{C}).$$

Recall the Abel map for the degree two divisor $D_\mu = \hat{\mu}_1 + \hat{\mu}_2$ based at P_0 :

$$(4.16) \quad \alpha_{P_0}(D_\mu) = \mathcal{A}_{P_0}(\hat{\mu}_1) + \mathcal{A}_{P_0}(\hat{\mu}_2) = \left(\int_{P_0}^{\hat{\mu}_1} w_1 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_1 d\lambda, \int_{P_0}^{\hat{\mu}_1} w_2 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_2 d\lambda \right).$$

Let $\{w_1 d\lambda, w_2 d\lambda\}$ be a basis of holomorphic differentials normalized by

$$(4.17) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad i, j = 1, 2.$$

On the affine chart write each differential as a rational function of λ divided by $y(\lambda)$:

$$(4.18) \quad w_i(\lambda) = (d_{i0} + d_{i1}\lambda) \frac{1}{y(\lambda)}, \quad d_{ik} \in \mathbb{C}, \quad i = 1, 2, \quad k = 0, 1.$$

Thus

$$w_1(\lambda) = \frac{d_{10} + d_{11}\lambda}{y(\lambda)}, \quad w_2(\lambda) = \frac{d_{20} + d_{21}\lambda}{y(\lambda)}.$$

Differentiate the components of $\alpha_{P_0}(D_\mu)$ with respect to x . By the chain rule, for $j = 1, 2$ and $i = 1, 2$,

$$(4.19) \quad \frac{d}{dx} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,x} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,x}.$$

We compute the w_1 -component of $\alpha_{P_0}(D_\mu)$. Using (4.19) and the expressions (3.38) and (3.45) for $\mu_{j,x}$ in terms of the Lax matrix entries (so that $\mu_{j,x} = \frac{2L_{11}(\mu_j)}{\mu_j - \mu_k}$ with $k \neq j$ and $y(\mu_j) = iL_{11}(\mu_j)$), one obtains

$$\begin{aligned} \frac{d}{dx} \left(\int_{P_0}^{\hat{\mu}_1} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda \right) &= \frac{d_{10} + d_{11}\mu_1}{y(\mu_1)} \mu_{1,x} + \frac{d_{10} + d_{11}\mu_2}{y(\mu_2)} \mu_{2,x} \\ &= d_{10} \left(\frac{\mu_{1,x}}{y(\mu_1)} + \frac{\mu_{2,x}}{y(\mu_2)} \right) + d_{11} \left(\frac{\mu_1 \mu_{1,x}}{y(\mu_1)} + \frac{\mu_2 \mu_{2,x}}{y(\mu_2)} \right) \\ &= d_{10} \left(\frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2} \frac{1}{iL_{11}(\mu_1)} + \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1} \frac{1}{iL_{11}(\mu_2)} \right) + d_{11} \left(\frac{2\mu_1}{i(\mu_1 - \mu_2)} + \frac{2\mu_2}{i(\mu_2 - \mu_1)} \right) \\ &= \frac{2}{i} d_{10} \left(\frac{1}{\mu_1 - \mu_2} + \frac{1}{\mu_2 - \mu_1} \right) + \frac{2}{i} d_{11} \left(\frac{\mu_1}{\mu_1 - \mu_2} + \frac{\mu_2}{\mu_2 - \mu_1} \right) \\ (4.20) \quad &= \frac{2}{i} d_{11} = -2i d_{11}. \end{aligned}$$

The w_2 -component is analogous. Repeating the same substitutions gives

$$(4.21) \quad \frac{d}{dx} \left(\int_{P_0}^{\hat{\mu}_1} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda \right) = -2i d_{21}.$$

Combining (4.20) and (4.21) we obtain

$$(4.22) \quad \frac{d}{dx} (\alpha_{P_0}(D_\mu)) = -2i (d_{11}, d_{21}) = -i \hat{U},$$

where $\hat{U} = (U_1, U_2)$ and $U_i = \oint_{b_i} d\Omega_2$.

For the time evolution induced by the higher flow one finds similarly

$$(4.23) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = -i \hat{V} = -i v \hat{U} = -2i v (d_{11}, d_{21}),$$

with

$$v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2), \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

Supplementary computations:

Differentiate the components of $\alpha_{P_0}(D_\mu)$ with respect to t . By the chain rule, for $j = 1, 2$ and $i = 1, 2$,

$$(4.24) \quad \frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,t} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,t}.$$

Use the evolution law (3.56) (and the analogous formula for $\mu_{2,t}$):

$$(4.25) \quad \mu_{j,t} = \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)}, \quad k \neq j.$$

Also use $y(\mu_j) = iL_{11}(\mu_j)$. Substituting these into (4.24) gives, for each j ,

$$\frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = \frac{d_{i0} + d_{i1}\mu_j}{iL_{11}(\mu_j)} \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)} = \frac{2}{iu} \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j).$$

Summing the contributions from $j = 1, 2$ we obtain the two components of the time derivative of $\alpha_{P_0}(D_\mu)$. Thus

$$(4.26) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} \left(\sum_{j=1}^2 \frac{d_{10} + d_{11}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j), \sum_{j=1}^2 \frac{d_{20} + d_{21}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) \right),$$

with in each sum $k \neq j$.

Using the relation $L_{12} = \frac{1}{4}V_{12} + \alpha_1 u$ from (2.18), and noting that $L_{12}(\mu_j) = 0$ for $j = 1, 2$, we obtain

$$(4.27) \quad V_{12}(\mu_j) = 4(L_{12}(\mu_j) - \alpha_1 u) = -4\alpha_1 u, \quad j = 1, 2.$$

Hence for each component $i \in \{1, 2\}$,

$$(4.28) \quad \sum_{j=1}^2 \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) = -4\alpha_1 u \left(\frac{d_{i0} + d_{i1}\mu_1}{\mu_1 - \mu_2} + \frac{d_{i0} + d_{i1}\mu_2}{\mu_2 - \mu_1} \right) = -4\alpha_1 u \cdot d_{i1}.$$

Substituting (4.28) into (4.26) gives

$$(4.29) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} (-4\alpha_1 u) (d_{11}, d_{21}) = 8i \alpha_1 (d_{11}, d_{21}).$$

From $\det L = -L_{11}^2 - L_{12}L_{21}$ and (2.26) write

$$L_{11} = -i(\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2)), \quad L_{12} = u(\lambda^2 + a\lambda + b), \quad L_{21} = -u(\lambda^2 - a\lambda + b),$$

with $a = i\frac{u_x}{2u}$ and $b = -\frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2$. Compute the leading terms in λ :

$$-L_{11}^2 = (\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2))^2 = \lambda^6 + 2(\alpha_1 - \frac{1}{2}u^2)\lambda^4 + \mathcal{O}(\lambda^2),$$

and

$$-L_{12}L_{21} = u^2(\lambda^4 + \mathcal{O}(\lambda^2)) = u^2\lambda^4 + \mathcal{O}(\lambda^2).$$

Hence

$$\det L = \lambda^6 + [2(\alpha_1 - \frac{1}{2}u^2) + u^2]\lambda^4 + \mathcal{O}(\lambda^2) = \lambda^6 + 2\alpha_1\lambda^4 + \mathcal{O}(\lambda^2).$$

On the other hand (2.28) gives

$$\det L = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = \lambda^6 - \left(\sum_{j=1}^3 \lambda_j^2 \right) \lambda^4 + \mathcal{O}(\lambda^2).$$

Equating the λ^4 -coefficients yields $\alpha_1 = -\frac{1}{2} \sum_{j=1}^3 \lambda_j^2$.

Week 5: Algebraic-Geometric Solutions for the mKdV Equation (5)

2025-10-21

Objectives: Express ω_2 , ω_3 , and $\hat{U}$ via the Riemann bilinear relations, and obtain the identities relating $\alpha_{P_0}(D_{\mu}^*)$ and $\alpha_{P_0}(D_{\mu})$ from Abel's theorem.

Riemann bilinear relation.

(i) Let $\hat{\Omega}$ and $\tilde{\Omega}$ be (multi-valued) Abelian integrals on the Riemann surface R_2 , and write their corresponding differentials as $d\hat{\Omega}$ and $d\tilde{\Omega}$. For the standard canonical homology basis $\{a_i, b_i\}_{i=1}^2$ the following identities hold by elementary contour rearrangements:

$$(5.1) \quad \oint_{a_i} \hat{\Omega} d\tilde{\Omega} + \oint_{a_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{a_i} \left(\int_{z_0}^z d\hat{\Omega} \right) d\tilde{\Omega} + \oint_{a_i^{-1}} \left(\int_{z_0}^{z'} d\hat{\Omega} \right) d\tilde{\Omega}$$

$$(5.2) \quad = - \oint_{a_i} \left(\int_z^{z'} d\hat{\Omega} \right) d\tilde{\Omega} = - \left(\oint_{a_i} d\hat{\Omega} \right) \left(\oint_{b_i} d\tilde{\Omega} \right),$$

$$(5.3) \quad \oint_{b_i} \hat{\Omega} d\tilde{\Omega} + \oint_{b_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{b_i} \left(\int_{z'}^z d\hat{\Omega} \right) d\tilde{\Omega} = \left(\oint_{a_i} d\hat{\Omega} \right) \left(\oint_{b_i} d\tilde{\Omega} \right).$$

Applying Stokes' theorem to the boundary of the canonical polygon for R_2 gives

$$(5.4) \quad 0 = \int_{R_2} d(\hat{\Omega} d\tilde{\Omega}) = \int_{\partial R_2} \hat{\Omega} d\tilde{\Omega} = \sum_{i=1}^2 \left(\oint_{a_i} + \oint_{b_i} + \oint_{a_i^{-1}} + \oint_{b_i^{-1}} \right) \hat{\Omega} d\tilde{\Omega},$$

and hence the usual bilinear identity

$$(5.5) \quad \sum_{i=1}^2 \left(\oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) = 0.$$

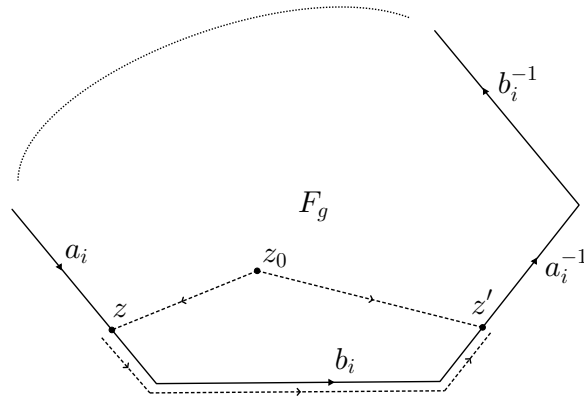


FIGURE 2. To the proof of the Riemann bilinear relations.

(ii) If the product $\hat{\Omega} d\tilde{\Omega}$ has poles (only) at the points $\infty^\pm$, then performing the same computation on $R_2 \setminus \{\infty^\pm\}$ introduces the boundary integral around small loops encircling the poles. One obtains

$$(5.6) \quad \sum_{i=1}^2 \left(\oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) + \oint_{l+\pi(l)} \hat{\Omega} d\tilde{\Omega} = 0,$$

where $l + \pi(l)$ denotes the small contours around the poles at $\infty^\pm$ with the orientation induced by the boundary of $R_2 \setminus \{\infty^\pm\}$.

Applications. Recall the Abelian integrals Ω_2, Ω_3 from (4.8)–(4.9). Define

$$\hat{U} = (U_1, U_2), \quad U_i = \oint_{b_i} d\Omega_2, \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

(1) The constants ω_2, ω_3 . Choose

$$\hat{\Omega} = \Omega_2, \quad d\tilde{\Omega} = \frac{\lambda^2}{y} d\lambda.$$

Since $\oint_{a_i} d\Omega_2 = 0$ and $\oint_{b_i} d\Omega_2 = U_i$, the bilinear identity (5.6) becomes

$$(5.7) \quad - \sum_{i=1}^2 U_i \oint_{a_i} \frac{\lambda^2}{y} d\lambda + \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 0.$$

The boundary integral is the sum of residues at $\infty^\pm$:

$$(5.8) \quad \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 2\pi i \left(\text{Res}_{\lambda=\infty^+} \Omega_2 \frac{\lambda^2}{y} d\lambda + \text{Res}_{\lambda=\infty^-} \Omega_2 \frac{\lambda^2}{y} d\lambda \right).$$

Using (4.8) and the large- λ expansion of y , one obtains

$$(5.9) \quad \Omega_2 \frac{\lambda^2}{y} = (\lambda + \omega_2 + o(1)) \lambda^{-1} \left(1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = 1 + \frac{\omega_2}{\lambda} + \dots$$

as $P \rightarrow \infty^\pm$. Thus the residue at each of $\infty^\pm$ equals ω_2 .

Substitution of (5.8) and (5.9) into (5.7) yields

$$(5.10) \quad \omega_2 = -\frac{i}{4\pi} \sum_{j=1}^2 U_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

Repeating the computation with $\hat{\Omega} = \Omega_3$ gives

$$(5.11) \quad \omega_3 = -\frac{i}{4\pi} \sum_{j=1}^2 V_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

(2) The vector $\hat{U}$. We now verify that

$$\hat{U} = (U_1, U_2) = (2d_{11}, 2d_{21}), \quad U_1 = 2d_{11}.$$

Choose $\hat{\Omega} = \Omega_2$ and

$$\tilde{\Omega} = w_1 d\lambda = (d_{10} + d_{11}\lambda) y^{-1} d\lambda, \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}.$$

Applying the bilinear identity gives

$$\sum_{i=1}^2 \left(\oint_{a_i} d\Omega_2 \oint_{b_i} w_1 d\lambda - \oint_{b_i} d\Omega_2 \oint_{a_i} w_1 d\lambda \right) + 2\pi i (\text{Res}_{\infty^+} \Omega_2 w_1 d\lambda + \text{Res}_{\infty^-} \Omega_2 w_1 d\lambda) = 0.$$

To determine the residues, use the large- λ expansions of Ω_2 and y :

$$\Omega_2 w_1 = (\lambda + \omega_2 + o(1))(d_{10} + d_{11}\lambda) \lambda^{-3} \left(1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = \frac{d_{11}}{\lambda} + \dots$$

as $P \rightarrow \infty^\pm$. Hence each of the points $\infty^\pm$ contributes a residue d_{11} , and the total equals $2d_{11}$. Since

$$\oint_{a_i} d\Omega_2 = 0, \quad \oint_{b_i} d\Omega_2 = U_i, \quad \oint_{a_i} w_1 d\lambda = 2\pi i \delta_{1i},$$

the bilinear identity reduces to

$$-2\pi i U_1 + 2\pi i (2d_{11}) = 0, \quad \text{so} \quad U_1 = 2d_{11}.$$

The same computation with w_2 gives $U_2 = 2d_{21}$.

Abel's theorem. Let $M(R_2)$ denote the meromorphic functions on R_2 , and $M'(R_2)$ the meromorphic 1-forms. For $f \in M(R_2) \setminus \{0\}$ and $P \in R_2$, the order $\text{ord}_P(f)$ is the integer k such that, in a local coordinate z with $z(P) = 0$,

$$f(z) = z^k g(z), \quad g(0) \neq 0.$$

The divisor of f is

$$(f) = \sum_{P \in R_2} \text{ord}_P(f) P,$$

which is a finite sum because f has only finitely many zeros and poles. The divisor (w) of a meromorphic 1-form $w \in M'(R_2) \setminus \{0\}$ is defined analogously by using its local expression $w(z) = z^k h(z) dz$ with $h(0) \neq 0$.

A meromorphic function is holomorphic if and only if $(f) \geq 0$. A meromorphic 1-form is holomorphic (of the first kind) if and only if $(w) \geq 0$. A divisor of the form (f) is called principal; a divisor of the form (w) is canonical.

Theorem 1 (Abel's theorem). *A divisor D is principal if and only if $\deg D = 0$ and $\alpha_{P_0}(D) = 0$.*

As an example,

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*} - D_{\infty^+, \mu},$$

which has degree zero. Abel's theorem then gives

$$\alpha_{P_0}(D_{\mu^*}) + \mathcal{A}_{P_0}(\infty^-) = \alpha_{P_0}(D_{\infty^-, \mu^*}) = \alpha_{P_0}(D_{\infty^+, \mu}) = \alpha_{P_0}(D_\mu) + \mathcal{A}_{P_0}(\infty^+),$$

hence

$$\alpha_{P_0}(D_{\mu^*}) = \alpha_{P_0}(D_\mu) + \mathcal{A}_{\infty^-}(\infty^+),$$

where

$$\mathcal{A}_{P_0}(P) = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

Week 6: Algebraic-Geometric Solutions for the mKdV Equation (6)

2025-10-28

Objectives: Focus on Case 2. Determine the homology basis under the involution τ , and compute the conjugate quantities $\overline{\ln(\omega_1)}$, $\overline{\omega_2}$, $\overline{\omega_3}$, and $\overline{B_{ij}}$.

Recall the definition (3.3):

$$(6.1) \quad R_2 := \left\{ (\lambda, y) \mid y^2 = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2) \right\}, \quad P = (\lambda, y).$$

We work in **Case 2** (one nonreal quadruple and one purely imaginary pair). Introduce the involutions

$$(6.2) \quad \tau_1 : (\lambda, y) \mapsto (\lambda, -y), \quad \tau : (\lambda, y) \mapsto (\bar{\lambda}, \bar{y}),$$

where τ_1 is the hyperelliptic (sheet-exchange) involution and τ is the antiholomorphic involution; both satisfy

$$\tau_1^2 = \tau^2 = \mathbb{I}.$$

Fix a canonical homology basis $\{a_1, a_2; b_1, b_2\}$ on R_2 with the intersection properties

$$(6.3) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij}, \quad i, j = 1, 2.$$

The antiholomorphic involution τ acts by sign change on the a -cycles and acts on the b -cycles by a combination of a - and b -cycles. Write

$$(6.4) \quad \tau(a_i) = -a_i, \quad \tau(b_j) = \sum_{k=1}^2 \alpha_{jk} a_k + \sum_{k=1}^2 \beta_{jk} b_k, \quad j = 1, 2,$$

for some integers α_{jk}, β_{jk} . Using the intersection pairing (6.3) we determine these coefficients as follows.

Compute the intersections with a_1, a_2 :

$$a_1 \circ \tau(b_1) = \beta_{11}(a_1 \circ b_1) = \beta_{11}, \quad a_2 \circ \tau(b_1) = \beta_{12}(a_2 \circ b_2) = \beta_{12}.$$

But the geometry of the antiholomorphic involution in this configuration yields

$$a_1 \circ \tau(b_1) = 1, \quad a_2 \circ \tau(b_1) = 0,$$

hence

$$(6.5) \quad \beta_{11} = 1, \quad \beta_{12} = 0.$$

Similarly, intersecting $\tau(b_1)$ with the b -cycles gives

$$\tau(b_1) \circ b_1 = \alpha_{11} = 0, \quad \tau(b_1) \circ b_2 = \alpha_{12} = 1,$$

so that $\alpha_{11} = 0, \alpha_{12} = 1$. Collecting these values yields

$$(6.6) \quad \tau(b_1) = b_1 + a_2.$$

By the same type of computation one obtains

$$(6.7) \quad \tau(b_2) = b_2 + a_1.$$

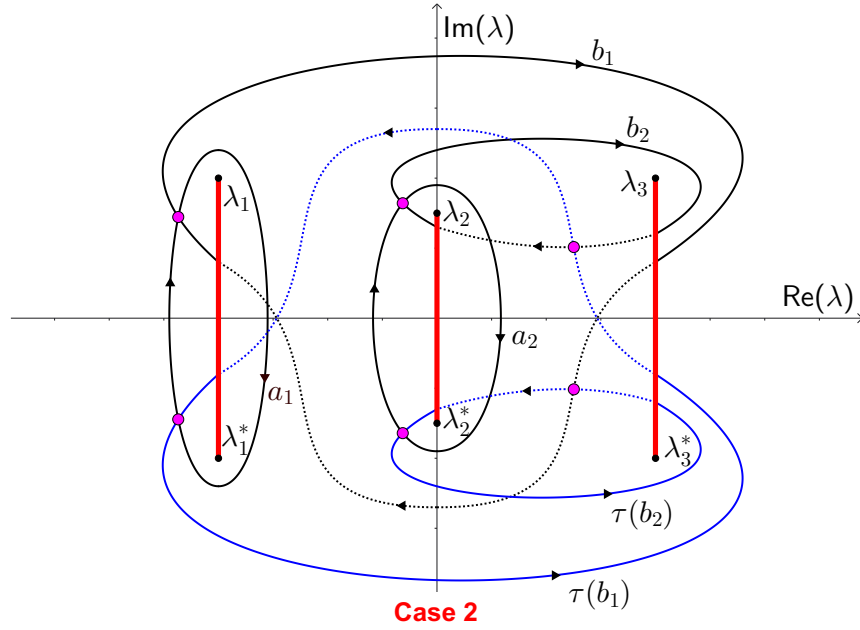


FIGURE 3. To the computations of $\tau(b_1), \tau(b_2)$ for Case 2.

Denote by $P_{\pm}$ the two points above the same λ -value on the two sheets of R_2 . Let C_P be the oriented path which starts at P_- (on the second sheet), crosses the cut joining the branch points λ_3, λ_3^* through their midpoint, then continues on the first sheet and winds counterclockwise along the cut until it reaches P_+ . (This is the path used below to connect ∞^- to ∞^+ in the indicated way.)

The path C_P is transformed by τ as

$$(6.8) \quad \tau(C_P) = C_P + a_3,$$

since $\tau(C_P) - C_P = a_3$. Computing the intersections $a_3 \circ b_j$ from the geometry one finds

$$a_3 \circ b_1 = -1, \quad a_3 \circ b_2 = -1,$$

and therefore

$$(6.9) \quad a_3 = -a_1 - a_2.$$

In fact one must include the small loop around ∞^+ . Thus the correct homology relation is

$$(6.10) \quad a_3 = -a_1 - a_2 + \ell,$$

where ℓ denotes a small positively oriented loop encircling ∞^+ .

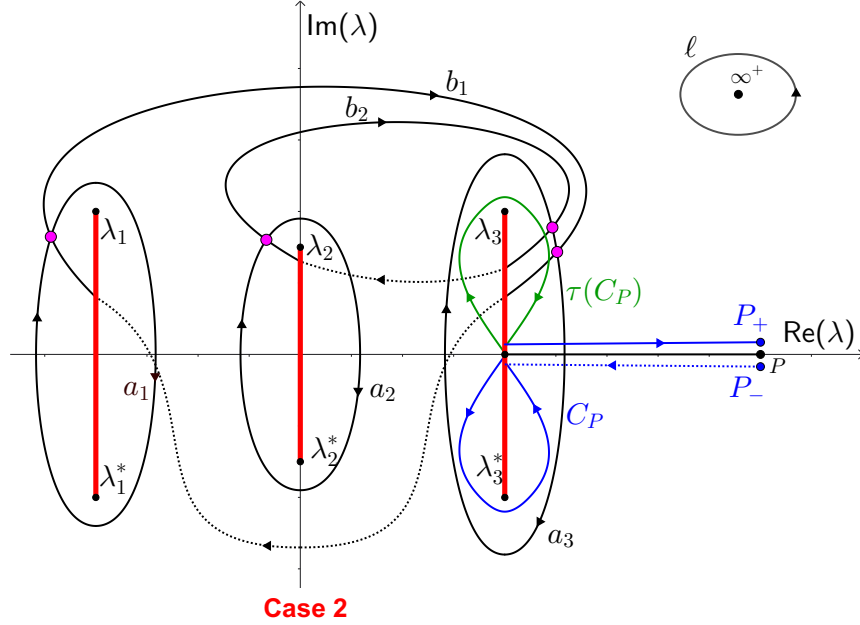


FIGURE 4. To the computation of $\tau(C_P)$ for Case 2.

Let $d\Omega_1$ be the normalized (third-kind) Abelian differential whose a -periods vanish:

$$(6.11) \quad \oint_{a_i} d\Omega_1 = 0, \quad i = 1, 2,$$

and which has the prescribed logarithmic behaviour at the points at infinity:

$$(6.12) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1 = \pm (\ln \lambda + \ln \omega_1 + o(1)), \quad P \rightarrow \infty^\pm,$$

where ω_1 is the (nonzero) constant appearing in the local expansion and w_i will denote holomorphic differentials as before.

Since $d\Omega_1$ is normalized by $\oint_{a_i} d\Omega_1 = 0$ and has residue 1 at ∞^+ , we obtain

$$(6.13) \quad \oint_{\ell} d\Omega_1 = 2\pi i.$$

Therefore

$$(6.14) \quad \oint_{a_3} d\Omega_1 = \oint_{-a_1-a_2+\ell} d\Omega_1 = -\sum_{i=1}^2 \oint_{a_i} d\Omega_1 + \oint_{\ell} d\Omega_1 = 0 + 2\pi i = 2\pi i.$$

Using the path C_P joining ∞^- to ∞^+ and the relation $\tau(C_P) = C_P + a_3$, the asymptotics (6.12) imply that

$$(6.15) \quad \lim_{P \rightarrow \infty^+} \left(\int_{\infty^-}^{\infty^+} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_1 - 2 \ln \lambda \right) = 2 \ln \omega_1.$$

We compute the conjugate quantity

$$(6.16) \quad \begin{aligned} 2 \overline{\ln(\omega_1)} &= \lim_{P \rightarrow \infty^+} \left(\int_{C_P} \overline{d\Omega_1(P)} - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{\tau(C_P)} d\Omega_1 - 2 \ln \lambda \right) \\ &= \lim_{P \rightarrow \infty^+} \left(\int_{C_P + a_3} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_1 - 2 \ln \lambda \right) + \oint_{a_3} d\Omega_1 \\ &= 2 \ln(\omega_1) + 2\pi i, \end{aligned}$$

where in the second equality we used $\overline{d\Omega_1(P)} = \tau^* d\Omega_1$ and $\tau(C_P) = C_P + a_3$, and in the last step we used (6.14). From (6.16) we obtain

$$(6.17) \quad \overline{\ln(\omega_1)} = \ln(\omega_1) + \pi i, \quad \Rightarrow \quad \Im(\ln(\omega_1)) = -\frac{\pi}{2}.$$

Hence, writing ω_1 in modulus-argument form gives

$$(6.18) \quad \ln(\omega_1) = \ln|\omega_1| - i\beta, \quad \beta > 0,$$

(and in our normalization one finds $\beta = \frac{\pi}{2}$).

The other normalized Abelian integrals have the asymptotics

$$(6.19) \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

so that, using the same connecting path C_P ,

$$(6.20) \quad 2\omega_2 = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_2 - 2\lambda \right), \quad 2\omega_3 = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_3 - 8\lambda^3 \right).$$

Taking complex conjugates and using the action of τ on C_P one deduces

$$(6.21) \quad 2\overline{\omega_2} = 2\omega_2, \quad 2\overline{\omega_3} = 2\omega_3, \quad \Rightarrow \quad \omega_2, \omega_3 \in \mathbb{R}.$$

Recall the period matrix entries are defined by

$$(6.22) \quad B_{ij} = \oint_{b_j} w_i d\lambda,$$

where $\{w_1, w_2\}$ denote the normalized holomorphic differentials with

$$\oint_{a_k} w_i d\lambda = 2\pi i \delta_{ik}, \quad i, k = 1, 2.$$

Using the antiholomorphic involution τ and the relations $\tau(b_1) = b_1 + a_2$, $\tau(b_2) = b_2 + a_1$ we get

$$(6.23) \quad \begin{aligned} \overline{B_{ij}} &= \oint_{b_j} \overline{w_i d\lambda} = \oint_{\tau(b_j)} w_i d\lambda = \oint_{b_j} w_i d\lambda + \sum_{l \neq j} \oint_{a_l} w_i d\lambda \\ &= B_{ij} + 2\pi i \sum_{l \neq j} \delta_{il}. \end{aligned}$$

Therefore, in matrix form,

$$(6.24) \quad \overline{B} = B + 2\pi i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The Riemann theta function is defined by

$$(6.25) \quad \Theta(z | B) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top z\right).$$

Using (6.24) one checks the conjugation property

$$(6.26) \quad \overline{\Theta(z \mid B)} = \Theta(\bar{z} \mid \bar{B}) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top \bar{z} + 2\pi i n_1 n_2\right) = \Theta(\bar{z} \mid B),$$

since $\exp(2\pi i n_1 n_2) = 1$ for integers n_1, n_2 .

Notational conventions and auxiliary differentials. For later calculations we introduce the following integrals and differentials:

$$(6.27) \quad \mathcal{Y}_{a_i}^{[j]} = \oint_{a_i} \frac{\lambda^j}{y} d\lambda, \quad \mathcal{Y}_{b_i}^{[j]} = \oint_{b_i} \frac{\lambda^j}{y} d\lambda, \quad i = 1, 2, \quad j \geq 0,$$

$$(6.28) \quad \Omega_j(P) = \int_{(\lambda_3, 0)}^P d\Omega_j, \quad d\Omega_2^{(1)} = \sum_{i=0}^3 \frac{c_{2i} \lambda^i}{y(\lambda)} d\lambda, \quad d\Omega_3^{(1)} = \sum_{i=1}^5 \frac{c_{3i} \lambda^i}{y(\lambda)} d\lambda,$$

where $P = (\lambda, y) \in R_2$, the base point $(\lambda_3, 0)$ is fixed, and the constants c_{2i}, c_{3i} (as well as the coefficients d_{ik} appearing in the holomorphic differentials) will be determined later on.

Week 7: Algebro-Geometric Solutions for the mKdV Equation (7)

2025-11-04

Objectives: Focus on Case 1. Introduce two transformations used in the computation of $\mathcal{Y}_{a_i}^{[j]}$ and $\mathcal{Y}_{b_i}^{[j]}$; carry out the evaluation of $\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{a_i}^{[1]}$, and determine the coefficients d_{ij} .

Recall that

$$w_i(\lambda) = \frac{d_{i0} + d_{i1}\lambda}{y(\lambda)}, \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

and

$$(7.1) \quad \oint_{a_j} w_i(\lambda) d\lambda = 2\pi i \delta_{ij}, \quad i, j = 1, 2.$$

Hence, writing the period integrals in matrix form, the normalization (7.1) is equivalent to

$$(7.2) \quad 2\pi i \mathbb{I}_2 = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}.$$

In particular,

$$(7.3) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = 2\pi i \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

Recall the intersection and period conventions for the canonical homology basis $\{a_i, b_i\}_{i=1}^2$:

$$(7.4) \quad a_i \circ a_j = b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij},$$

$$(7.5) \quad \oint_{b_j} w_i(\lambda) d\lambda = B_{ij}.$$

Case 1. $\lambda_1, \lambda_2, \lambda_3 \in i\mathbb{R}$.

Let τ denote complex conjugation on the curve,

$$\tau: (\lambda, y) \mapsto (\lambda^*, y^*).$$

The action of τ on the homology basis may be chosen (up to orientation) as

$$(7.6) \quad \tau(a_1) = -a_2, \quad \tau(a_2) = -a_1, \quad \tau(b_1) = b_2, \quad \tau(b_2) = b_1.$$

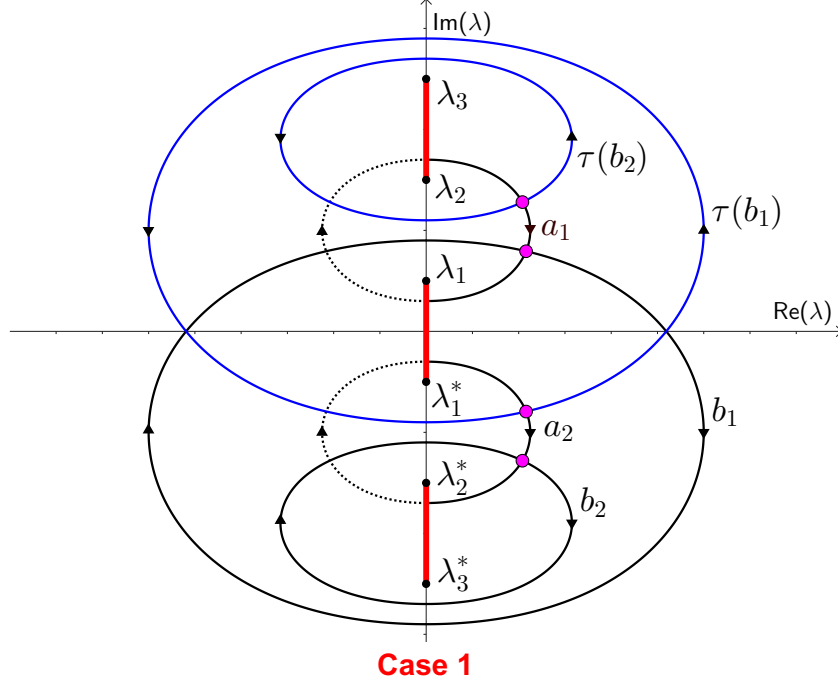


FIGURE 5. To the computations of $\tau(b_1), \tau(b_2)$ for Case 1.

With the choice of cycles described above, the a -periods reduce to integrals along paths on the first sheet:

$$(7.7) \quad \oint_{a_1} \frac{d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y}, \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y},$$

and

$$(7.8) \quad \oint_{a_2} \frac{d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{d\lambda}{y}, \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{\lambda d\lambda}{y}.$$

Recall the definition

$$(7.9) \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}, \quad \Lambda = \lambda^2, \quad d\Lambda = 2\lambda d\lambda.$$

Applying the substitution $\Lambda = \lambda^2$ transforms the basic integral into a rational function of Λ :

$$(7.10) \quad \int \frac{d\lambda}{y(\lambda)} = \int \frac{d\lambda}{\sqrt{\prod_{i=1}^3 (\lambda^2 - \lambda_i^2)}} = \int \frac{d\Lambda}{2\sqrt{\Lambda \prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Thus the problem reduces to integrals of the general form

$$(7.11) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

To reduce (7.11) to the Legendre normal form we use the change of variable

$$(7.12) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}.$$

A direct computation yields the standard reduction

$$(7.13) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k^2 z^2)}},$$

with the elliptic modulus given by

$$(7.14) \quad k^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

For reference, the incomplete elliptic integrals in Jacobi form are

$$F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad E(z, k) = \int_0^z \sqrt{\frac{1-k^2t^2}{1-t^2}} dt, \quad 0 < k < 1,$$

and the elliptic integral of the third kind is

$$\Pi(z, \alpha^2, k) = \int_0^z \frac{dt}{(1-\alpha^2t^2)\sqrt{(1-t^2)(1-k^2t^2)}}.$$

These standard forms allow the evaluation of the a -period integrals (7.7)–(7.8) once the correspondence between the branch points $\{\beta_i\}$ and the original λ -branch points is fixed.

Applying (7.13)–(7.14) to our concrete problem: we may take

$$\beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

and (using (7.12)) obtain

$$(7.15) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\Lambda}{\Lambda - \lambda_1^2} = \frac{(\lambda_1^2 - \lambda_3^2) \lambda^2}{\lambda_3^2 (\lambda_1^2 - \lambda^2)}.$$

For these choices the modulus becomes

$$(7.16) \quad k_1^{(1)2} = \frac{(\lambda_1^2 - \lambda_2^2)(0 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{\lambda_3^2}{\lambda_2^2} \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2},$$

so that

$$(7.17) \quad \int \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_1^{(1)2}z^2)}}, \quad k_1^{(1)} = \frac{\lambda_3}{\lambda_2} \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

Defining the incomplete elliptic integral

$$(7.18) \quad v = F(z, k_1^{(1)}) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_1^{(1)2}t^2)}}, \quad z = \text{sn}(v, k_1^{(1)}),$$

we obtain the standard elliptic parametrisation of λ via (7.15) and (7.18).

Recall the relations:

$$\text{sn}^2(u, k) + \text{cn}^2(u, k) = 1, \quad \text{dn}^2(u, k) + k^2 \text{sn}^2(u, k) = 1,$$

and the complete elliptic integrals

$$K(k) = F(1, k), \quad K'(k) = F(1, k'), \quad k'^2 = 1 - k^2.$$

From (7.15) one checks the mapping of special Λ -values to z -values:

$$\Lambda = \lambda^2 = 0 \leftrightarrow z = 0 \leftrightarrow v = 0, \quad \Lambda = \lambda_1^2 \leftrightarrow z^2 \rightarrow \infty \leftrightarrow v = \pm iK_1^{(1)'}$$

and

$$\Lambda = \lambda^2 = \lambda_2^2 \leftrightarrow z^2 = \frac{\lambda_2^2}{\lambda_3^2} \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda_2^2} = \frac{1}{k_1^{(1)2}} \leftrightarrow v = \pm (K_1^{(1)} + iK_1^{(1)'}).$$

With this choice, the a_1 -period can be expressed in the v -variable as an integral between two points in the complex v -plane. In particular, we have

$$(7.19) \quad 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1^{(1)} + iK_1^{(1)'}}^{iK_1^{(1)'}} dv = \frac{-2K_1^{(1)}}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} = \frac{-2iK_1^{(1)'}}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-.$$

The symmetry under complex conjugation, $\tau : (\lambda, y) \mapsto (\lambda^*, y^*)$, implies the relations

$$(7.20) \quad \left(\oint_{a_2} \frac{d\lambda}{y} \right)^* = \oint_{a_2} \tau^* \frac{d\lambda}{y} = \oint_{\tau(a_2)} \frac{d\lambda}{y} = \oint_{-a_1} \frac{d\lambda}{y} = - \oint_{a_1} \frac{d\lambda}{y} \in \mathbb{R}^+.$$

Similarly, one finds

$$(7.21) \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = \oint_{a_1} \frac{\lambda d\lambda}{y}.$$

Using the substitution $\Lambda = \lambda^2$, one rewrites

$$(7.22) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \int \frac{d\Lambda}{2\sqrt{\prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Introducing the change of variables

$$(7.23) \quad z^2 = \frac{\beta_1 - \beta_3}{\Lambda - \beta_3},$$

the integral becomes

$$(7.24) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\lambda_1^2 - \lambda_3^2}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_2^{(1)2}z^2)}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}.$$

Hence,

$$(7.25) \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = \frac{-2i K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}} = \oint_{a_2} \frac{\lambda d\lambda}{y}.$$

Since

$$(7.26) \quad \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

the relation (7.3) becomes

$$\begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

This yields the coefficients

$$(7.27) \quad \begin{aligned} d_{10} &= -\frac{\pi \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_1^{(1)}}, & d_{11} &= -\frac{\pi \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_2^{(1)}}, \\ d_{20} &= -d_{10}, & d_{21} &= d_{11}. \end{aligned}$$

Supplementary computations:

Define

$$(7.28) \quad F(z, k) = \int_0^z \frac{dt}{f(t, k)}, \quad f(t, k) = \sqrt{(1-t^2)(1-k^2t^2)}, \quad 0 < k < 1,$$

where the principal branch of the square root is chosen and we fix $f(0, k) = 1$.

Set

$$(7.29) \quad K(k) = F(1, k), \quad K'(k) = K(k'), \quad k' = \sqrt{1-k^2}.$$

We take the segments $[-k^{-1}, -1] \cup [1, k^{-1}]$ as branch cuts for $f(t, k)$ on the extended real axis. Starting at $t = 0$ and moving right along the real axis, each time one crosses a branch point an extra phase factor $e^{-\pi i/2} = -i$ is accumulated; moving left from 0 produces the conjugate factor $e^{\pi i/2} = i$. With these conventions, $f(t, k)$ on the first sheet admits the explicit real-axis description

$$(7.30) \quad f(t, k) = \begin{cases} \sqrt{(1-t^2)(1-k^2t^2)} \in \mathbb{R}^+, & t \in (-1, 1), \\ -i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^-, & t \in (1, k^{-1}), \\ i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^+, & t \in (-k^{-1}, -1), \\ -\sqrt{(t^2-1)(k^2t^2-1)} \in \mathbb{R}^-, & t \in (-\infty, -k^{-1}) \cup (k^{-1}, \infty). \end{cases}$$

We now evaluate several definite integrals that will be used below.

For the interval (k^{-1}, ∞) perform the substitution

$$(7.31) \quad t = k^{-1}s^{-1}, \quad dt = -k^{-1}s^{-2}ds,$$

which maps $t \in [k^{-1}, \infty)$ to $s \in (0, 1]$. Direct computation gives

$$(7.32) \quad \begin{aligned} \int_{k^{-1}}^{\infty} \frac{dt}{\sqrt{(t^2-1)(k^2t^2-1)}} &= \int_1^0 \frac{-k^{-1}s^{-2}ds}{\sqrt{(k^{-2}s^{-2}-1)(s^{-2}-1)}} \\ &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-k^2s^2)}} \\ &= F(1, k) = K(k). \end{aligned}$$

For the finite interval $(1, k^{-1})$ use the substitution

$$(7.33) \quad t = (1 - (k')^2s^2)^{-1/2}, \quad dt = (k')^2s(1 - (k')^2s^2)^{-3/2}ds,$$

which sends $s \in [0, 1]$ to $t \in [1, k^{-1}]$. Under this change of variables

$$(7.34) \quad \begin{aligned} \sqrt{(t^2-1)(1-k^2t^2)} &= \sqrt{\frac{(k's)^2}{1-(k's)^2} \cdot (k')^2 \frac{1-s^2}{1-(k's)^2}} \\ &= (k')^2s \frac{\sqrt{1-s^2}}{1-(k's)^2}. \end{aligned}$$

Hence

$$(7.35) \quad \begin{aligned} \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-(k')^2s^2)}} \\ &= F(1, k') = K(k') = K'(k). \end{aligned}$$

Combining the evaluations on $[0, 1]$ and $(1, k^{-1})$, and recalling the phase $-i$ on $(1, k^{-1})$ from (7.30), we obtain

$$(7.36) \quad \begin{aligned} F(k^{-1}, k) &= \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= K(k) + iK'(k). \end{aligned}$$

Taking the limit $z \rightarrow +\infty$ along the positive real axis (passing through k^{-1}) yields

$$(7.37) \quad F(+\infty, k) = K(k) + iK'(k) - K(k) = iK'(k),$$

while symmetry across the origin gives

$$(7.38) \quad F(-1, k) = -F(1, k) = -K(k).$$

Similarly, evaluating F at $-k^{-1}$ (follow the path from 0 to -1 to $-k^{-1}$ and use the phase convention on $(-k^{-1}, -1)$ in (7.30)) yields

$$(7.39) \quad \begin{aligned} F(-k^{-1}, k) &= -K(k) + \int_{-1}^{-k^{-1}} \frac{dt}{i\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + iK'(k). \end{aligned}$$

Finally, taking $z \rightarrow -\infty$ along the negative real axis (passing through $-k^{-1}$) gives

$$(7.40) \quad F(-\infty, k) = -K(k) + iK'(k) + K(k) = iK'(k).$$

Since

$$(type\ 1) \quad \int \frac{\lambda^{2n}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2(\Lambda(\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

and

$$(type\ 2) \quad \int \frac{\lambda^{2n+1}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2((\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

we treat the two types of integrals separately.

Computation of Type 1 Integrals $(\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{b_i}^{[0]})$. Introduce a change of variable from Λ to z by

$$(7.41) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

Solving (7.41) for Λ yields

$$(7.42) \quad \Lambda = \frac{z^2\beta_3(\beta_4 - \beta_1) - (\beta_3 - \beta_1)\beta_4}{z^2(\beta_4 - \beta_1) - (\beta_3 - \beta_1)} = \beta_3 + \frac{(\beta_3 - \beta_1)(\beta_3 - \beta_4)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$

Differentiation gives

$$(7.43) \quad d\Lambda = \frac{2(\beta_3 - \beta_1)(\beta_4 - \beta_3)(\beta_4 - \beta_1)z}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} dz.$$

The shifted factors $\Lambda - \beta_i$ become rational functions of z . In particular

$$(7.44) \quad \Lambda - \beta_1 = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(z^2 - 1)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.45) \quad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.46) \quad \Lambda - \beta_4 = \frac{(\beta_3 - \beta_4)(\beta_4 - \beta_1)z^2}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$

Hence the full quartic factor factors as

$$(7.47) \quad \prod_{i=1}^4 (\Lambda - \beta_i) = \frac{(\beta_3 - \beta_1)^2 (\beta_4 - \beta_1)^2 (\beta_4 - \beta_3)^2 z^2 (z^2 - 1)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^4} \\ \times \left((\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2) \right).$$

Define the modulus

$$(7.48) \quad k_1^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

Taking a principal square root of (7.47) (and grouping algebraic constants) yields

$$(7.49) \quad \left(\prod_{i=1}^4 (\Lambda - \beta_i) \right)^{1/2} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} \sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}.$$

Combining (7.43) and (7.49) produces the standard elliptic reduction

$$(7.50) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_4 - \beta_2)(\beta_3 - \beta_1)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_1^2 z^2)}},$$

valid on the appropriate choice of branches for the square roots.

Specializing the parameters by

$$(7.51) \quad \beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

the change of variable (7.41) becomes

$$(7.52) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\lambda^2}{\lambda^2 - \lambda_1^2}, \quad z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}},$$

where the positive square root is taken for the principal branch. The modulus (7.48) simplifies to

$$(7.53) \quad k_1^2 = \frac{(\lambda_1^2 - \lambda_2^2)(\lambda_2^2 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2}.$$

Choose the cut $[-\lambda_1, \lambda_1]$ for the function $\sqrt{\lambda^2 - \lambda_1^2}$ and fix the sheet by requiring

$$\sqrt{\lambda^2 - \lambda_1^2} \sim \lambda \quad (\lambda \rightarrow \infty).$$

For $\text{Re } \lambda \geq 0$ and $\lambda \in i\mathbb{R}$ write $\lambda = it$ with $t \in \mathbb{R}$. If $\lambda_j = it_j$ then

$$(7.54) \quad \sqrt{\lambda^2 - \lambda_1^2} = \begin{cases} i\sqrt{t^2 - t_1^2}, & t > t_1, \\ \sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -i\sqrt{t^2 - t_1^2}, & t < -t_1. \end{cases}$$

Hence for $\lambda \in i\mathbb{R}$ on the imaginary axis with $\text{Re } \lambda \geq 0$,

$$(7.55) \quad z = \begin{cases} \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}} \in i\mathbb{R}, & \lambda \in (-\lambda_1, \lambda_1), \\ \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{|\lambda|}{\sqrt{\lambda_1^2 - \lambda^2}} \in \mathbb{R}^+, & \lambda \in (-i\infty, -\lambda_1) \cup (\lambda_1, +i\infty). \end{cases}$$

The oriented real/imaginary segments map under $\lambda \mapsto z$ as follows (the images are indicated in the z -plane):

$$(7.56) \quad \begin{aligned} (\lambda_3, \lambda_2) &\longrightarrow (1, k_1^{-1}), & (\lambda_2, \lambda_1) &\longrightarrow (k_1^{-1}, +\infty), \\ (\lambda_1, 0) &\longrightarrow (+i\infty, 0), & (0, -\lambda_1) &\longrightarrow (0, -i\infty), \\ (-\lambda_1, -\lambda_2) &\longrightarrow (+\infty, k_1^{-1}), & (-\lambda_2, -\lambda_3) &\longrightarrow (k_1^{-1}, 1). \end{aligned}$$

Since

$$(7.57) \quad \lambda y(\lambda) = \sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)},$$

for our chosen map between λ and z we may write

$$\lambda y(\lambda) = c_0 z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}, \quad c_0 > 0.$$

For $\lambda = it \in i\mathbb{R}$, note that $z \in (0, 1)$ when $\lambda \in (\lambda_3, +i\infty)$; in this region $\lambda y(\lambda) \in \mathbb{R}^+$. The branch points $\pm\lambda_3, \pm\lambda_2$ of $y(\lambda)$ correspond to the branch points $1, k_1^{-1}$ of $f(z, k_1)$, where $f(z, k_1) = \sqrt{(1 - z^2)(1 - k_1^2 z^2)}$ and the branch cuts are chosen as in (7.28)–(7.30).

Thus, for $\lambda \in i\mathbb{R}$,

$$\lambda y(\lambda) \sim \begin{cases} z f(z, k_1), & \lambda \in (-\lambda_2, +i\infty) \cup (-\lambda_3, -i\infty), \\ -z f(z, k_1), & \lambda \in (-\lambda_2, -\lambda_3). \end{cases}$$

Hence for $\lambda \in i\mathbb{R} \setminus [-\lambda_2, -\lambda_3]$,

$$(7.58) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{f(z, k_1)},$$

and on the interval $(-\lambda_2, -\lambda_3)$, where the sign of the square root changes,

$$(7.59) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{-f(z, k_1)}.$$

Using (7.58)–(7.59) together with the standard elliptic integrals, and noting

$$(7.60) \quad \sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)} = -i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2} > 0,$$

one obtains the following closed forms

$$(7.61) \quad \begin{aligned} \mathcal{Y}_{a_1}^{[0]} &= 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^{+\infty} \frac{dz}{f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1 + iK_1'}^{iK_1'} dv = \frac{-2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-, \end{aligned}$$

$$(7.62) \quad \mathcal{Y}_{a_2}^{[0]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{+\infty}^{k_1^{-1}} \frac{dz}{f(z, k_1)} = -\mathcal{Y}_{a_1}^{[0]} = \frac{2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.63) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^1 \frac{dz}{-f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1}^{K_1 + iK_1'} dv = \frac{-2K_1'}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^+, \end{aligned}$$

and

$$\begin{aligned}
 \mathcal{Y}_{b_1}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{d\lambda}{y(\lambda)} = -\frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} + \frac{2(-iK'_1 - iK'_1)}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \\
 (7.64) \qquad \qquad \qquad &= \frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-.
 \end{aligned}$$

Computation of Type 2 Integrals $(\mathcal{Y}_{a_i}^{[1]}, \mathcal{Y}_{b_i}^{[1]})$.

Let $\beta_1 < \beta_2 < \beta_3$ and introduce the substitution

$$(7.65) \qquad z^2 = \frac{\beta_1 - \beta_2}{\Lambda - \beta_3}, \quad \Lambda = \beta_3 + \frac{\beta_1 - \beta_3}{z^2}, \quad \Lambda = \lambda^2.$$

Differentiation yields

$$(7.66) \qquad d\Lambda = \frac{2(\beta_3 - \beta_1)}{z^3} dz.$$

Using (7.65) one checks

$$(7.67) \qquad \Lambda - \beta_1 = (\beta_3 - \beta_1) \frac{z^2 - 1}{z^2},$$

$$(7.68) \qquad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)z^2 + (\beta_1 - \beta_3)}{z^2},$$

$$(7.69) \qquad \Lambda - \beta_3 = \frac{\beta_1 - \beta_3}{z^2}.$$

Consequently the cubic product factors as

$$(7.70) \qquad \prod_{i=1}^3 (\Lambda - \beta_i) = (\beta_3 - \beta_1)^3 \frac{(z^2 - 1)(1 - k_2^2 z^2)}{z^6},$$

with

$$(7.71) \qquad k_2^2 = \frac{\beta_3 - \beta_2}{\beta_3 - \beta_1} \in (0, 1).$$

Taking principal square roots gives

$$(7.72) \qquad \left(\prod_{i=1}^3 (\Lambda - \beta_i) \right)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}.$$

Combining (7.66) and (7.72) we obtain the reduction

$$(7.73) \qquad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^3 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{\beta_1 - \beta_3}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}.$$

Specialize $\beta_1 = \lambda_3^2$, $\beta_2 = \lambda_2^2$, $\beta_3 = \lambda_1^2$. Then

$$(7.74) \qquad z^2 = \frac{\lambda_3^2 - \lambda_2^2}{\lambda^2 - \lambda_1^2}, \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}},$$

and

$$(7.75) \qquad k_2^2 = \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2} \in (0, 1).$$

Choose the cut $[-\lambda_1, \lambda_1]$ for $\sqrt{\lambda_1^2 - \lambda^2}$ and fix the sheet by the condition

$$\sqrt{\lambda_1^2 - \lambda^2} \sim -i\lambda \quad (\lambda \rightarrow \infty).$$

For $\lambda = it$ ($t \in \mathbb{R}$) and $\lambda_j = it_j$ one then has the pointwise values (for $\text{Re } \lambda \geq 0$)

$$(7.76) \quad \sqrt{\lambda_1^2 - \lambda^2} = \begin{cases} \sqrt{t^2 - t_1^2}, & t > t_1, \\ -i\sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -\sqrt{t^2 - t_1^2}, & t < -t_1, \end{cases}$$

which are consistent with the chosen asymptotics.

Consequently, for $\lambda \in i\mathbb{R}$ with $\text{Re } \lambda \geq 0$, the substitution takes values as follows

$$(7.77) \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}} \in \begin{cases} \mathbb{R}^+, & \lambda \in (\lambda_1, +i\infty), \\ i\mathbb{R}^+, & \lambda \in (-\lambda_1, \lambda_1), \\ \mathbb{R}^-, & \lambda \in (-i\infty, -\lambda_1). \end{cases}$$

The oriented segments in the λ -plane map to the z -plane under (7.77) as follows (principal images):

$$(7.78) \quad (+i\infty, \lambda_3) \longrightarrow (0, 1), \quad (\lambda_3, \lambda_2) \longrightarrow (1, k_2^{-1}), \quad (\lambda_2, \lambda_1) \longrightarrow (k_2^{-1}, +\infty),$$

$$(7.79) \quad (\lambda_1, 0) \longrightarrow \left(+i\infty, \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}\right), \quad (0, -\lambda_1) \longrightarrow \left(\frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}, +\infty\right),$$

$$(7.80) \quad (-\lambda_1, -\lambda_2) \longrightarrow (-\infty, -k_2^{-1}), \quad (-\lambda_2, -\lambda_3) \longrightarrow (-k_2^{-1}, -1), \quad (-\lambda_3, -i\infty) \longrightarrow (-1, 0).$$

By the Type 2 reduction (with $\Lambda = \lambda^2$) the cubic factor admits the representation

$$y(\lambda) = \prod_{i=1}^3 (\Lambda - \beta_i)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}, \quad \beta_j = \lambda_{4-j}^2,$$

On the chosen sheet for $\lambda \in i\mathbb{R}$ along the imaginary axis, comparison of signs yields

$$y(\lambda) = -i(\beta_3 - \beta_1) \sqrt{\beta_3 - \beta_1} \frac{f(z, k_2)}{z^3}, \quad f(z, k_2) = \sqrt{(1 - z^2)(1 - k_2^2 z^2)}.$$

where the branch cuts of $f(z, k_2)$ are chosen as in (7.28)–(7.30). Hence, for $\lambda \in i\mathbb{R}$,

$$\frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\beta_3 - \beta_1}} \frac{dz}{f(z, k_2)} = \frac{i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \frac{dz}{f(z, k_2)},$$

Introducing the complete elliptic integrals $K_2 = K(k_2)$ and $K'_2 = K(k'_2)$, and using the standard substitutions along the indicated images, the period integrals (type 2 moments with a factor λ) evaluate as follows.

$$(7.81) \quad \mathcal{Y}_{a_1}^{[1]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{k_2^{-1}}^{+\infty} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2 + iK'_2}^{iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.82) \quad \mathcal{Y}_{a_2}^{[1]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-\infty}^{-k_2^{-1}} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_2}^{-K_2 + iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.83) \quad \mathcal{Y}_{b_2}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-k_2^{-1}}^{-1} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-K_2 + iK'_2}^{-K_2} dv = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.84) \quad \mathcal{Y}_{b_1}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \mathcal{Y}_{b_2}^{[1]} = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+.$$

Week 8: Algebraic-Geometric Solutions for the mKdV Equation (8)

2025-11-11

Objectives: Focus on Case 1. Compute $\mathcal{Y}_{b_i}^{[0]}$, and discuss how to determine the value of a multivalued complex function with a prescribed branch cut, taking $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$ with the asymptotic condition $y(\lambda) \sim \lambda^3$ as $\lambda \rightarrow \infty$.

We have

$$(8.1) \quad B = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{b_1} \frac{d\lambda}{y} & \oint_{b_2} \frac{d\lambda}{y} \\ \oint_{b_1} \frac{\lambda d\lambda}{y} & \oint_{b_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_1}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

We can obtain

$$(8.2) \quad \int_0^\lambda \frac{\chi d\chi}{\sqrt{\prod_{i=1}^3 (\chi^2 - \lambda_i^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_2^{(1)2}t^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^v dv.$$

and

$$(8.3) \quad \lambda^2 = \frac{\lambda_3^2 - \lambda_1^2 \operatorname{cn}^2(v, k_2^{(1)})}{\operatorname{sn}^2(v, k_2^{(1)})}, \quad \operatorname{sn}(v, k_2^{(1)}) = z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

and

$$v = F(z, k_2^{(1)}) = F\left(\sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, k_2^{(1)}\right)$$

Supplementary computations:

Let $P = (\lambda, y)$ be a point on the Riemann surface defined by

$$y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

The first sheet is fixed by the asymptotic condition

$$y(\lambda) \sim \lambda^3 + O(\lambda), \quad \lambda \rightarrow \infty.$$

Set

$$(8.4) \quad \lambda_j = i t_j, \quad 0 < t_1 < t_2 < t_3, \quad j = 1, 2, 3.$$

For the chosen cut we rewrite $y(\lambda)$ for $\lambda = it \in i\mathbb{R}$ (keeping the factorised form with the fractional square roots):

$$(8.5) \quad \begin{aligned} y(\lambda) &= (\lambda - \lambda_3) \sqrt{\frac{\lambda - \lambda_2}{\lambda - \lambda_3}} (\lambda - \lambda_1) \sqrt{\frac{\lambda + \lambda_1}{\lambda - \lambda_1}} (\lambda + \lambda_2) \sqrt{\frac{\lambda + \lambda_3}{\lambda + \lambda_2}} \\ &= -i(t - t_3)(t - t_1)(t + t_2) \sqrt{\frac{t - t_2}{t - t_3}} \sqrt{\frac{t + t_1}{t - t_1}} \sqrt{\frac{t + t_3}{t + t_2}}. \end{aligned}$$

Here the square root is taken as the principal branch:

$$(8.6) \quad \sqrt{z} = \sqrt{|z|} e^{\frac{1}{2}i \arg z}, \quad \arg z \in (-\pi, \pi].$$

To determine the permutation of real/imaginary values and the signs of $y(it)$ as t varies along the imaginary axis, it is convenient to count

- the sign $s = \text{sign}((t - t_3)(t - t_1)(t + t_2))$ of the real linear prefactor, and
- the number $k \in \{0, 1, 2, 3\}$ of the three real ratios

$$\frac{t - t_2}{t - t_3}, \quad \frac{t + t_1}{t - t_1}, \quad \frac{t + t_3}{t + t_2}$$

that are negative (each negative ratio contributes a factor i from the principal square root).

With these conventions, from (8.5) one has

$$y(it) = (-i)^s (i)^k (\text{positive real}) = -s i^{k+1} (\text{positive real}),$$

so that $y(it)$ is imaginary when $k \equiv 0, 2 \pmod{2}$ and real when $k \equiv 1, 3 \pmod{2}$, with the sign determined by the combination above.

Evaluating s and k on the natural partition of the real axis by $\pm t_1, \pm t_2, \pm t_3$ gives the following classification:

$$(8.7) \quad y(it) \in \begin{cases} i\mathbb{R}^-, & t > t_3, \\ \mathbb{R}^-, & t_2 < t < t_3, \\ i\mathbb{R}^+, & t_1 < t < t_2, \\ \mathbb{R}^+, & -t_1 < t < -t_2, \\ i\mathbb{R}^-, & -t_2 < t < -t_1, \\ \mathbb{R}^-, & -t_3 < t < -t_2, \\ i\mathbb{R}^+, & t < -t_3. \end{cases}$$

For clarity, a short justification for each interval is as follows (sign of linear prefactor s and number k of negative ratios are given in parentheses):

- $t > t_3$: $s = +1, k = 0 \Rightarrow y = -i \cdot (+1) \cdot 1 \cdot (\text{pos}) \in i\mathbb{R}^-$.
- $t_2 < t < t_3$: $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$.
- $t_1 < t < t_2$: $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$.
- $-t_1 < t < -t_2$: $s = +1, k = 1 \Rightarrow y = +(\text{pos}) \in \mathbb{R}^+$.
- $-t_2 < t < -t_1$: $s = +1, k = 0 \Rightarrow y = -i \cdot (\text{pos}) \in i\mathbb{R}^-$.
- $-t_3 < t < -t_2$: $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$.
- $t < -t_3$: $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$.

All statements above are obtained without algebraically combining the fractional square-root factors; each square root is treated as the principal branch (8.6) applied to the linear fractional arguments appearing in (8.5).

Near each branch point λ_j or $\lambda_j^* = -\lambda_j$ the function behaves locally as $\sqrt{\lambda \mp \lambda_j}$ (analytic part), and starting from $\lambda = -i\infty$ (where $y \in i\mathbb{R}^+$), moving upward along the right side of the imaginary axis one acquires at each branch point an additional phase $i\pi/2$, equivalently a multiplicative factor i , fully consistent with the above sign computation.

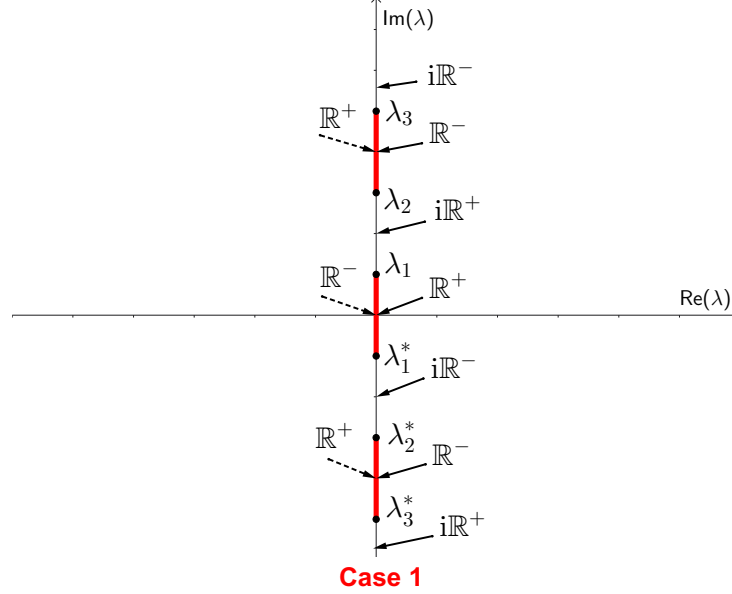


FIGURE 6. The sign values of $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$, with $\lambda \rightarrow \infty$, $y \rightarrow \lambda^3 + O(\lambda)$ on the first sheet for Case 1.

Week 9: Algebraic-Geometric Solutions for the mKdV Equation (9)

2025-11-18

Objectives: Focus on Case 1. Introduce the Joukowski function to parametrize the two transformations used in the computation of $\mathcal{Y}_{a_i}^{[j]}$; compute $\mathcal{Y}_{b_i}^{[1]}$ and B_{ij} ; and discuss how to determine a suitable ansatz for the differentials $d\Omega_j$ for $j = 1, 2, 3$.

Introduce the Joukowski transformation

$$(9.1) \quad \lambda = \frac{\lambda_1}{2}(\xi + \xi^{-1}).$$

A direct computation gives

$$(9.2) \quad \lambda_1^2 - \lambda^2 = \lambda_1^2 \left(1 - \left(\frac{1}{2}(\xi + \xi^{-1}) \right)^2 \right) = - \left(\frac{\lambda_1}{2} \right)^2 (\xi - \xi^{-1})^2.$$

For a type 2 integral (see (7.74)),

$$(9.3) \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}} = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{i \frac{\lambda_1}{2}} \frac{1}{\xi - \xi^{-1}}.$$

For a type 1 integral (see (7.52)), using (9.1) one obtains

$$(9.4) \quad z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\frac{\lambda_1}{2}(\xi + \xi^{-1})}{\frac{\lambda_1}{2}(\xi - \xi^{-1})} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\xi^2 + 1}{\xi^2 - 1}.$$

As $\lambda \rightarrow \infty$, the function y admits the expansion

$$(9.5) \quad y = \lambda^3 \left(1 + \hat{v}_1 \lambda^{-2} + \hat{v}_2 \lambda^{-4} + \dots \right), \quad y^{-1} = \lambda^{-3} \left(1 - \hat{v}_1 \lambda^{-2} + (\hat{v}_1^2 - \hat{v}_2) \lambda^{-4} + \dots \right).$$

Decompose y as

$$(9.6) \quad y = y^{[n+]} + y^{[n-]} = \lambda^3 \left(1 + \hat{v}_1 \lambda^{-2} + \dots + \hat{v}_n \lambda^{-2n} \right) + y^{[n-]},$$

where $y^{[n-]} = \hat{v}_{n+1} \lambda^{-2(n+1)+3} + \dots$. Multiplying (9.6) by y^{-1} and using (9.5) gives

$$(9.7) \quad y^{[n+]} y^{-1} = (y - y^{[n-]}) y^{-1} = 1 - y^{[n-]} y^{-1}$$

$$(9.8) \quad = 1 - (\hat{v}_{n+1} \lambda^{-2(n+1)} + \dots) \lambda^3 \cdot \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots)$$

$$(9.9) \quad = 1 - \hat{v}_{n+1} \lambda^{-2(n+1)} + \dots.$$

In particular,

$$(9.10) \quad \lambda^3 y^{-1} = 1 - \hat{v}_1 \lambda^{-2} + \dots,$$

and

$$(9.11) \quad \lambda^3 (1 + \hat{v}_1 \lambda^{-2}) y^{-1} = 1 - \hat{v}_2 \lambda^{-4} + \dots,$$

with analogous identities holding for higher-order terms.

Since

$$(9.12) \quad \Omega_2(P) = \pm (\lambda + \omega_2 + o(1)), \quad p \rightarrow \infty^\pm,$$

differentiation shows that

$$(9.13) \quad d\Omega_2(P) = \pm (1 + O(\lambda^{-2})) d\lambda.$$

Thus $d\Omega_2$ can be written in the form

$$d\Omega_2(P) = \frac{1}{y} (\lambda^3 + c_{21} \lambda + c_{20}) d\lambda.$$

Supplementary computations:

The period normalization for the holomorphic differentials

$$w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}$$

is expressed by

$$2\pi i \delta_{ij} = \oint_{a_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{a_j}^{[k]}, \quad B_{ij} = \oint_{b_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{b_j}^{[k]}.$$

The symmetry properties of the period integrals,

$$\mathcal{Y}_{a_1}^{[0]} = -\mathcal{Y}_{a_2}^{[0]}, \quad \mathcal{Y}_{b_1}^{[0]} = -\mathcal{Y}_{b_2}^{[0]}, \quad \mathcal{Y}_{a_1}^{[1]} = \mathcal{Y}_{a_2}^{[1]}, \quad \mathcal{Y}_{b_1}^{[1]} = \mathcal{Y}_{b_2}^{[1]},$$

allow the normalization conditions to be written as

$$(9.14) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{a_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{a_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2\pi i \mathbb{I}_2,$$

and

$$(9.15) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = B^{(1)}.$$

Eliminating the coefficients d_{ik} between (9.14) and (9.15) gives

$$\begin{aligned}
(9.16) \quad B^{(1)} &= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\
&= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\
&= \pi i \begin{pmatrix} \tau_1^{(1)} & \tau_2^{(1)} \\ -\tau_1^{(1)} & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \pi i \begin{pmatrix} \tau_1^{(1)} + \tau_2^{(1)} & \tau_2^{(1)} - \tau_1^{(1)} \\ \tau_2^{(1)} - \tau_1^{(1)} & \tau_1^{(1)} + \tau_2^{(1)} \end{pmatrix}.
\end{aligned}$$

Here the reduced moduli are given by

$$(9.17) \quad \tau_1^{(1)} = (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} = i \frac{K_1^{(1)'}}{K_1^{(1)}}, \quad \tau_2^{(1)} = (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} = i \frac{K_2^{(1)'}}{K_2^{(1)}}.$$

By the definition of the Riemann theta function,

$$\begin{aligned}
(9.18) \quad \Theta(B^{(1)}; z) &= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{1}{2} n^\top B^{(1)} n + n^\top z \right\} \\
&= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{\pi i}{2} n^\top \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} n + n^\top z \right\},
\end{aligned}$$

where

$$(9.19) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad n_1, n_2 \in \mathbb{Z}, \quad z_1, z_2 \in \mathbb{C}.$$

Decompose n as

$$(9.20) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} + \begin{pmatrix} t \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix}, \quad t = 0, 1,$$

so that

$$(9.21) \quad n_1 = m_1 + m_2 + t = 2m_1 + n_2 + t, \quad n_2 = m_2 - m_1, \quad m_1, m_2 \in \mathbb{Z}.$$

As (m_1, m_2) range over $\mathbb{Z}^2$, so do (n_1, n_2) .

Substituting (9.20) into (9.18) yields

$$\begin{aligned}
(9.22) \quad \Theta(B^{(1)}; z) &= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left(m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix} \right. \\
&\quad \left. + \left(m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right\} \\
&= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left(m_1 + \frac{t}{2} \right)^2 \tau_1^{(1)} + 2\pi i \left(m_2 + \frac{t}{2} \right)^2 \tau_2^{(1)} \right. \\
&\quad \left. + \left(m_1 + \frac{t}{2} \right) (z_1 - z_2) + \left(m_2 + \frac{t}{2} \right) (z_1 + z_2) \right\} \\
&= \vartheta_3(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_3(z_1 + z_2, 2\tau_2^{(1)}) + \vartheta_2(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_2(z_1 + z_2, 2\tau_2^{(1)}).
\end{aligned}$$

Definition 3. The Jacobi theta functions are defined by

$$\begin{aligned}
(9.23) \quad \vartheta_1(u, \tau) &:= \Theta \begin{bmatrix} 1 \\ 1 \end{bmatrix} (u, \tau), \quad \vartheta_2(u, \tau) := \Theta \begin{bmatrix} 1 \\ 0 \end{bmatrix} (u, \tau), \\
\vartheta_3(u, \tau) &:= \Theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (u, \tau), \quad \vartheta_4(u, \tau) := \Theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u, \tau),
\end{aligned}$$

with $\text{Im } \tau > 0$ and

$$(9.24) \quad \Theta \left[\begin{smallmatrix} \epsilon \\ \epsilon' \end{smallmatrix} \right] (u, \tau) := \sum_{n \in \mathbb{Z}} \exp \left\{ \pi i \left(n + \frac{\epsilon}{2} \right)^2 \tau + \left(n + \frac{\epsilon}{2} \right) (u + \pi i \epsilon') \right\}, \quad \epsilon, \epsilon' \in \{0, 1\}.$$

Assume

$$(9.25) \quad d\Omega_2(P) = \sum_{i=0}^3 c_{2i} \frac{\lambda^i}{y(\lambda)} d\lambda, \quad d\Omega_3(P) = \sum_{i=0}^5 c_{3i} \frac{\lambda^i}{y(\lambda)} d\lambda.$$

The Abelian integrals satisfy, as $P = (\lambda, y) \rightarrow \infty^\pm$,

$$(9.26) \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)),$$

and differentiation yields the asymptotics of the differentials

$$(9.27) \quad d\Omega_2(P) = \pm(1 + O(\lambda^{-2})) d\lambda, \quad d\Omega_3(P) = \pm(12\lambda^2 + O(\lambda^{-2})) d\lambda.$$

Two equivalent ways determine the coefficients c_{2i} and c_{3i} .

Method A — matching via the asymptotic identities. Substitute the ansatzes (9.25) into the asymptotic identities (9.27) and use (9.10)-(9.11).

For $d\Omega_2$ we require

$$\left(\sum_{i=0}^3 c_{2i} \lambda^i \right) y^{-1} = \pm(1 + O(\lambda^{-2})) = \lambda^3 y^{-1},$$

hence

$$(9.28) \quad c_{23} = 1, \quad c_{22} = 0.$$

For $d\Omega_3$ the asymptotic condition reads

$$\left(\sum_{i=0}^5 c_{3i} \lambda^i \right) y^{-1} = \pm(12\lambda^2 + O(\lambda^{-2})) = 12\lambda^2(1 + O(\lambda^{-4})) = 12\lambda^2 \cdot \lambda^3(1 + \hat{v}_1 \lambda^{-2}) y^{-1},$$

and matching powers of λ gives

$$(9.29) \quad c_{35} = 12, \quad c_{34} = 0, \quad c_{33} = 12\hat{v}_1, \quad c_{32} = 0.$$

Method B — direct series expansion. Expand the rational expressions in powers of λ^{-1} using the explicit series (9.5) and compare coefficients.

For $d\Omega_2$ compute

$$(9.30) \quad \begin{aligned} \left(\sum_{i=0}^3 c_{2i} \lambda^i \right) y^{-1} &= (c_{23} \lambda^3 + c_{22} \lambda^2 + c_{21} \lambda + c_{20}) \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots) \\ &= c_{23} + c_{22} \lambda^{-1} + (c_{21} - c_{23} \hat{v}_1) \lambda^{-2} + \dots \end{aligned}$$

Matching with $\pm(1 + O(\lambda^{-2}))$ shows directly that $c_{23} = 1$ and $c_{22} = 0$, in agreement with (9.28).

For $d\Omega_3$ compute

$$(9.31) \quad \begin{aligned} \left(\sum_{i=0}^5 c_{3i} \lambda^i \right) y^{-1} &= (c_{35} \lambda^5 + c_{34} \lambda^4 + c_{33} \lambda^3 + c_{32} \lambda^2 + \dots) \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots) \\ &= c_{35} \lambda^2 + c_{34} \lambda + (c_{33} - \hat{v}_1 c_{35}) + (c_{32} - \hat{v}_1 c_{34}) \lambda^{-1} + \dots \end{aligned}$$

Matching with $(12\lambda^2 + O(\lambda^{-2}))$ yields $c_{35} = 12$, $c_{34} = 0$, and $c_{33} - \hat{v}_1 c_{35} = 0$, $c_{32} - \hat{v}_1 c_{34} = 0$, hence

$$c_{35} = 12, \quad c_{34} = 0, \quad c_{33} = 12\hat{v}_1, \quad c_{32} = 0,$$

which coincides with (9.29).

Thus both routes produce the same values for the top coefficients. Collecting the results gives the following forms

$$(9.32) \quad d\Omega_2(P) = \frac{\lambda^3 + c_{21} \lambda + c_{20}}{y(\lambda)} d\lambda,$$

and

$$(9.33) \quad d\Omega_3(P) = \frac{12\lambda^5 + 12\hat{v}_1\lambda^3 + c_{31}\lambda + c_{30}}{y(\lambda)} d\lambda,$$

where the remaining coefficients $c_{21}, c_{20}, c_{31}, c_{30}$ are fixed by the a -period normalization conditions.

Week 10: Algebro-Geometric Solutions for the mKdV Equation (10)

2025-11-25

Objectives: Focus on Case 1. Compute $c_{2i}, c_{3,i}$ and $\Omega_j^{(1)}(P)$.

The meromorphic differentials take the form

$$(10.1) \quad \begin{aligned} d\Omega_2^{(1)}(P) &= \frac{\lambda^3 + c_{21}\lambda + c_{20}}{y(\lambda)} d\lambda, \\ d\Omega_3^{(1)}(P) &= \frac{12\lambda^5 - 3v\lambda^3 + c_{31}\lambda + c_{30}}{y(\lambda)} d\lambda, \end{aligned}$$

with

$$(10.2) \quad v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2).$$

The constants $c_{21}, c_{20}, c_{31}, c_{30}$ are determined by imposing vanishing of the a -periods,

$$(10.3) \quad \oint_{a_i} d\Omega_2^{(1)} = \oint_{a_i} d\Omega_3^{(1)} = 0, \quad i = 1, 2.$$

For $d\Omega_2^{(1)}$ the conditions (10.3) yield the linear system

$$(10.4) \quad \begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & \oint_{a_1} \frac{d\lambda}{y} \\ \oint_{a_2} \frac{\lambda}{y} d\lambda & \oint_{a_2} \frac{d\lambda}{y} \end{pmatrix} \begin{pmatrix} c_{21} \\ c_{20} \end{pmatrix} = - \begin{pmatrix} \oint_{a_1} \frac{\lambda^3}{y} d\lambda \\ \oint_{a_2} \frac{\lambda^3}{y} d\lambda \end{pmatrix} = - \oint_{a_1} \frac{\lambda^3}{y} d\lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Using the symmetry of the chosen a -cycles the period matrix on the left of (10.4) decomposes as

$$\begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & \oint_{a_1} \frac{d\lambda}{y} \\ \oint_{a_2} \frac{\lambda}{y} d\lambda & \oint_{a_2} \frac{d\lambda}{y} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & 0 \\ 0 & \oint_{a_1} \frac{d\lambda}{y} \end{pmatrix},$$

so the system (10.4) is readily solved. The solution is

$$(10.5) \quad \begin{pmatrix} c_{21} \\ c_{20} \end{pmatrix} = - \oint_{a_1} \frac{\lambda^3}{y} d\lambda \begin{pmatrix} \left(\oint_{a_1} \frac{\lambda}{y} d\lambda \right)^{-1} & 0 \\ 0 & \left(\oint_{a_1} \frac{d\lambda}{y} \right)^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = - \begin{pmatrix} \oint_{a_1} \frac{\lambda^3}{y} d\lambda \left(\oint_{a_1} \frac{\lambda}{y} d\lambda \right)^{-1} \\ 0 \end{pmatrix}.$$

Using the previously computed values of the a_1 -periods,

$$(10.6) \quad \mathcal{Y}_{a_1}^{[3]} = - \frac{2i(\lambda_3^2 K_2^{(1)} + (\lambda_1^2 - \lambda_3^2)E_2^{(1)})}{\sqrt{\lambda_1^2 - \lambda_3^2}}, \quad \mathcal{Y}_{a_1}^{[1]} = - \frac{2i K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}},$$

one obtains the closed form

$$(10.7) \quad c_{21} = \frac{(\lambda_3^2 - \lambda_1^2)E_2^{(1)}}{K_2^{(1)}} - \lambda_3^2, \quad c_{20} = 0.$$

The same procedure applied to $d\Omega_3^{(1)}$ (matching its a -periods to zero) gives

$$(10.8) \quad c_{30} = 0, \quad c_{31} = 4v_1 - \lambda_3^2 v - v(\lambda_1^2 - \lambda_3^2) \frac{E_2^{(1)}}{K_2^{(1)}},$$

where

$$(10.9) \quad v_1 = \lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2,$$

and v is given by (10.2). Thus the constants appearing in (10.1) are determined by (10.7) and (10.8).

Thus

$$(10.10) \quad \begin{aligned} \Omega_2^{(1)}(P) &= \int_{(\lambda_3, 0)}^P d\Omega_2^{(1)} = \int_{\lambda_3}^{\lambda} \frac{\mu^3 + \left((\lambda_3^2 - \lambda_1^2) \frac{E_2^{(1)}}{K_2^{(1)}} - \lambda_3^2 \right) \mu}{y(\mu)} d\mu \\ &= \frac{1}{i\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2^{(1)}}^v \left(\frac{\lambda_3^2 - \lambda_1^2}{\operatorname{sn}^2(u, k_2^{(1)})} + (\lambda_3^2 - \lambda_1^2) \frac{E_2^{(1)}}{K_2^{(1)}} - (\lambda_3^2 - \lambda_1^2) \right) du \\ &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \int_{K_2^{(1)}}^v \left(-\frac{1}{\operatorname{sn}^2(u, k_2^{(1)})} - \frac{E_2^{(1)}}{K_2^{(1)}} + 1 \right) du, \end{aligned}$$

where

$$(10.11) \quad v = F\left(\sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, k_2^{(1)}\right), \quad \operatorname{sn}^2(v, k_2^{(1)}) = \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}, \quad \lambda^2 = \frac{\lambda_3^2 - \lambda_1^2 \operatorname{cn}^2(v, k_2^{(1)})}{\operatorname{sn}^2(v, k_2^{(1)})}.$$

Define the Jacobi Zeta function by

$$(10.12) \quad Z(u, k) = \frac{\partial}{\partial u} \ln \Theta_4\left(\frac{i\pi u}{K(k)}\right) = \int_0^u (\operatorname{dn}^2(z, k) - \frac{E(k)}{K(k)}) dz,$$

where $K(k)$ and $E(k)$ are the complete elliptic integrals of the first and second kind.

Writing $k = k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}$ and denoting the elliptic argument by x , substitution into (10.10) gives

$$(10.13) \quad \begin{aligned} \Omega_2^{(1)}(P) &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \int_{K_2^{(1)}}^v \left(\operatorname{dn}^2(x, k) + k^2 \operatorname{sn}^2(x, k) - \frac{E_2^{(1)}}{K_2^{(1)}} - \frac{1}{\operatorname{sn}^2(x, k)} \right) dx \\ &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \left[Z(x, k) + \frac{\operatorname{cn}(x, k) \operatorname{dn}(x, k)}{\operatorname{sn}(x, k)} \right]_{x=K_2^{(1)}}^{x=v}. \end{aligned}$$

Since the boundary term at $x = K_2^{(1)}$ vanishes, (10.13) becomes

$$(10.14) \quad \Omega_2^{(1)}(P) = -i\sqrt{\lambda_1^2 - \lambda_3^2} \left(Z(v, k) + \frac{\operatorname{cn}(v, k) \operatorname{dn}(v, k)}{\operatorname{sn}(v, k)} \right).$$

Using (10.11), one has

$$\operatorname{sn}(v, k) = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, \quad \operatorname{cn}(v, k) = \sqrt{\frac{\lambda_3^2 - \lambda^2}{\lambda_1^2 - \lambda^2}}, \quad \operatorname{dn}(v, k) = \sqrt{\frac{\lambda_2^2 - \lambda^2}{\lambda_1^2 - \lambda^2}},$$

and then

$$(10.15) \quad \frac{\operatorname{cn}(v, k) \operatorname{dn}(v, k)}{\operatorname{sn}(v, k)} = \sqrt{\frac{(\lambda_3^2 - \lambda^2)(\lambda_2^2 - \lambda^2)}{(\lambda_1^2 - \lambda^2)(\lambda_1^2 - \lambda_3^2)}}.$$

Similarly,

$$(10.16) \quad \Omega_3^{(1)}(P) = \int_{\lambda_3}^{\lambda} \frac{12\mu^5 - 4v\mu^3 + 4v_1\mu}{y(\mu)} d\mu + v\Omega_2^{(1)}(P) = 4y + v\Omega_2^{(1)}(P),$$

where v and v_1 are defined by (10.2) and (10.9).

Furthermore,

$$(10.17) \quad d\Omega_1^{(1)} = \frac{\lambda^2 + c_{10}}{y(\lambda)} d\lambda, \quad c_{10} = -\lambda_1^2 \Pi\left(\frac{\lambda_2^2 - \lambda_1^2}{\lambda_2^2}, k_1^{(1)}\right) / K_1^{(1)},$$

and the corresponding elliptic parametrization is

$$(10.18) \quad \nu = F\left(\sqrt{\frac{\lambda_1^2(\lambda_1^2 - \lambda_3^2)}{\lambda_3^2(\lambda_1^2 - \lambda_2^2)}}, k_1^{(1)}\right), \quad \lambda^2 = \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(\nu, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(\nu, k_1^{(1)})}.$$

Hence

$$(10.19) \quad \Omega_1^{(1)}(P) = \int_{\lambda_3}^{\lambda} \frac{\mu^2 + c_{10}}{y(\mu)} d\mu.$$

One has

$$(10.20) \quad \Pi(u, \alpha^2, k) = \frac{\operatorname{sn}(a, k)}{\operatorname{cn}(a, k) \operatorname{dn}(a, k)} \left(\frac{1}{2} \ln \frac{\vartheta_1\left(\frac{i\pi(a+u)}{K}, \tau\right)}{\vartheta_1\left(\frac{i\pi(a-u)}{K}, \tau\right)} - u Z(a, k) \right), \quad \alpha = \frac{1}{\operatorname{sn}(a, k)}$$

and

$$(10.21) \quad \Omega_1^{(1)}(P) = \frac{-i}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1^{(1)}}^{\nu} \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(x, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(x, k_1^{(1)})} - \frac{\lambda_1^2}{K_1^{(1)}} \Pi\left(\frac{\lambda_2^2 - \lambda_1^2}{\lambda_2^2}, k_1^{(1)}\right) dx.$$

Supplementary computations:

Computation of Type 1 Integrals ($\mathcal{Y}_{a_i}^{[2]}, \mathcal{Y}_{b_i}^{[2]}$).

We introduce the auxiliary function Λ (cf. (7.42)) by

$$(10.22) \quad \Lambda(z) = \frac{\lambda_1^2 z^2}{z^2 - \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_3^2 z^2 + (\lambda_1^2 - \lambda_3^2)} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_1^2 - \lambda_3^2 (1 - z^2)}.$$

Recall the relation between z and the elliptic argument ν ,

$$(10.23) \quad \nu = F(z, k_1^{(1)}), \quad z = \operatorname{sn}(\nu, k_1^{(1)}).$$

We will use the imaginary-period shift identities for Jacobi elliptic functions,

$$(10.24) \quad \operatorname{sn}(\nu + iK_1', k_1^{(1)}) = \frac{1}{k_1^{(1)} \operatorname{sn}(\nu, k_1^{(1)})}, \quad \operatorname{cn}(\nu + iK_1', k_1^{(1)}) = -i \frac{\operatorname{dn}(\nu, k_1^{(1)})}{k_1^{(1)} \operatorname{sn}(\nu, k_1^{(1)})}.$$

Set

$$(10.25) \quad \nu = \nu_1 + iK_1',$$

so that $\operatorname{sn}(\nu, k_1^{(1)})$ and $\operatorname{cn}(\nu, k_1^{(1)})$ in (10.22) may be expressed in terms of ν_1 . Substituting $z = \operatorname{sn}(\nu, k_1^{(1)})$ into (10.22) and using (10.24) yields

$$(10.26) \quad \begin{aligned} \Lambda &= \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(\nu, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(\nu, k_1^{(1)})} = \frac{\lambda_1^2 \lambda_3^2}{\lambda_1^2 (k_1^{(1)})^2 \operatorname{sn}^2(\nu_1, k_1^{(1)}) + \lambda_3^2 \operatorname{dn}^2(\nu_1, k_1^{(1)})} \\ &= \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + (k_1^{(1)})^2 (\lambda_1^2 - \lambda_3^2) \operatorname{sn}^2(\nu_1, k_1^{(1)})}. \end{aligned}$$

Using the identity

$$(k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2},$$

the expression (10.26) can be rewritten in the equivalent forms

$$(10.27) \quad \Lambda = \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + \frac{\lambda_3^2}{\lambda_2^2} (\lambda_1^2 - \lambda_2^2) \operatorname{sn}^2(\nu_1, k_1^{(1)})} = \frac{\lambda_1^2}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \operatorname{sn}^2(\nu_1, k_1^{(1)})}.$$

The last form highlights that Λ becomes a simple rational function of $\text{sn}^2(\nu_1, k_1^{(1)})$, which is convenient for integrating.

We now compute $\mathcal{Y}_{a_1}^{[2]}$. Starting from the integral in (type 1) and changing variables to the elliptic parameter ν (with the shift $\nu = \nu_1 + iK_1'$) one obtains

$$(10.28) \quad \mathcal{Y}_{a_1}^{[2]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1 + iK_1'}^{iK_1'} \frac{d\nu}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \text{sn}^2(\nu - iK_1', k_1^{(1)})}.$$

The integrand is an elementary function of sn^2 , hence the integral is an incomplete elliptic integral of the third kind. Writing the integral in terms of $\Pi(\alpha_1^2, u, k)$, with characteristic $\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}$, modulus $k = k_1^{(1)}$, and upper limit u , we have

$$(10.29) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_1^2, u, k_1^{(1)}) \Big|_{u=K_1}^{u=0}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

Evaluating the difference of the incomplete elliptic integrals between $u = K_1$ and $u = 0$ gives the complete elliptic integral of the third kind $\Pi(\alpha_1^2, k)$. Therefore

$$(10.30) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{-2i\lambda_1^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_1^2, k_1^{(1)}), \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

By symmetry of the contour (or by the corresponding sign change in the integration limits), one obtains the companion relation

$$(10.31) \quad \mathcal{Y}_{a_2}^{[2]} = -\mathcal{Y}_{a_1}^{[2]}.$$

Similarly,

$$(10.32) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1}^{K_1 + iK_1'} \frac{d\nu}{1 - \alpha_1^2 \text{sn}^2(\nu - iK_1', k_1^{(1)})} \\ &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \Pi(\alpha_1^2, u, k_1^{(1)}) \Big|_{u=K_1 - iK_1'}^{u=K_1}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}. \end{aligned}$$

We use the standard Jacobi identities

$$(10.33) \quad \text{sn}(z + K, k) = \frac{\text{cn}(z, k)}{\text{dn}(z, k)}, \quad \text{sn}(iz, k) = i \frac{\text{sn}(z, k')}{\text{cn}(z, k')},$$

$$(10.34) \quad \text{cn}(iz, k) = \frac{1}{\text{cn}(z, k')}, \quad \text{dn}(iz, k) = \frac{\text{dn}(z, k')}{\text{cn}(z, k')},$$

which imply

$$(10.35) \quad \text{dn}(z - K_1', k_1') = \frac{k_1}{\text{dn}(z, k_1')}, \quad \text{sn}(iz + K_1, k) = \frac{1}{\text{dn}(z, k')}.$$

Hence,

$$(10.36) \quad \text{sn}(iu + K_1 - iK_1', k_1^{(1)}) = \frac{1}{\text{dn}(u - K_1', k_1^{(1)'})} = \frac{\text{dn}(u, k_1^{(1)'})}{k_1^{(1)}}.$$

Let $v = iu + K_1$. Then

$$\begin{aligned}
\mathcal{Y}_{b_2}^{[2]} &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \operatorname{sn}^2(iu + K_1 - iK'_1, k_1^{(1)})} \\
(10.37) \quad &= \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \left(\frac{1}{(k_1^{(1)})^2} - \left(\frac{k_1^{(1)'}}{k_1^{(1)}} \right)^2 \operatorname{sn}^2(u, k_1^{(1)'}) \right)} \\
&= \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{du}{1 - \alpha_2^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_2^2, k_1^{(1)'}), \quad \alpha_2^2 = \frac{\lambda_2^2 - \lambda_3^2}{\lambda_2^2}.
\end{aligned}$$

One has

$$\begin{aligned}
\mathcal{Y}_{b_1}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} \\
(10.38) \quad &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_1}^{-iK'_1} \frac{dv}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})}.
\end{aligned}$$

We simplify the elliptic integrand using standard shifts of Jacobi functions. For complementary modulus k' one has

$$\operatorname{sn}(u - K', k') = -\frac{\operatorname{cn}(u, k')}{\operatorname{dn}(u, k')}, \quad \operatorname{cn}(u - K', k') = \frac{k \operatorname{sn}(u, k')}{\operatorname{dn}(u, k')}.$$

From these identities it follows that,

$$\operatorname{sn}(iu - iK'_1, k_1^{(1)}) = i \frac{\operatorname{sn}(u - K'_1, k_1^{(1)'})}{\operatorname{cn}(u - K'_1, k_1^{(1)'})} = -\frac{i}{k_1^{(1)}} \frac{\operatorname{cn}(u, k_1^{(1)'})}{\operatorname{sn}(u, k_1^{(1)'})}.$$

Now set $v = iu$ and substitute the last expression into the integrand. Using

$$\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}, \quad (k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2)\lambda_3^2}{(\lambda_1^2 - \lambda_3^2)\lambda_2^2},$$

we compute the denominator appearing in the integrand

$$\begin{aligned}
1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)}) &= 1 - \frac{\lambda_1^2 - \lambda_2^2}{\lambda_2^2} \frac{1}{(k_1^{(1)})^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} = 1 - \frac{\lambda_1^2 - \lambda_3^2}{\lambda_3^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} \\
(10.39) \quad &= \frac{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2) \operatorname{cn}^2(u, k_1^{(1)'})}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

This expression for the denominator is used to rewrite the integral in (10.38) in terms of elliptic functions with argument u . From (10.39) we obtain the partial-fraction-type decomposition

$$\begin{aligned}
\frac{1}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})} &= \frac{\lambda_3^2}{\lambda_1^2} + \frac{\frac{\lambda_3^2}{\lambda_1^2} (\lambda_1^2 - \lambda_3^2)}{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)} \\
(10.40) \quad &= \frac{\lambda_3^2}{\lambda_1^2} - \frac{\lambda_3^2}{\lambda_1^2} \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

Hence the relation between $\mathcal{Y}_{b_1}^{[2]}$ and $\mathcal{Y}_{b_2}^{[2]}$ follows:

$$\begin{aligned}
(10.41) \quad \mathcal{Y}_{b_1}^{[2]} &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_3^2}{-i\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1'}^{-K_1'} \left(1 - \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})} \right) i du \\
&= \mathcal{Y}_{b_2}^{[2]} + \frac{4\lambda_3^2 K_1'}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2}, k_1^{(1)'}\right) \\
&= \mathcal{Y}_{b_2}^{[2]} + 2\lambda_3^2 \mathcal{Y}_{b_1}^{[0]} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\alpha_3^2, k_1^{(1)'}\right), \quad \alpha_3^2 = \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2},
\end{aligned}$$

where the middle line uses the definition of the complete elliptic integral of the third kind and the last line expresses the K_1' -term via $\mathcal{Y}_{b_1}^{[0]}$.

Computation of Type 2 Integrals ($\mathcal{Y}_{a_i}^{[3]}, \mathcal{Y}_{b_i}^{[3]}$).

For integrals of type 2, we use the parametrization

$$(10.42) \quad \Lambda = \lambda_1^2 + \frac{\lambda_3^2 - \lambda_1^2}{z^2}, \quad z = \operatorname{sn}(\nu, k_2^{(1)}), \quad (k_2^{(1)})^2 = \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda_3^2}.$$

Writing $\nu = \nu_1 + iK'$, this expression admits several equivalent forms. Indeed,

$$\Lambda = \lambda_1^2 + (\lambda_3^2 - \lambda_1^2) (k_2^{(1)})^2 \operatorname{sn}^2(\nu_1, k_2^{(1)}) = \lambda_1^2 + (\lambda_3^2 - \lambda_1^2) (1 - \operatorname{dn}^2(\nu_1, k_2^{(1)})),$$

and hence

$$\Lambda = \lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu_1, k_2^{(1)}).$$

Using this representation, we compute

$$(10.43) \quad \mathcal{Y}_{a_1}^{[3]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda^3 d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2 + iK_2'}^{iK_2'} \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu - iK_2', k_2^{(1)}) \right) d\nu.$$

The integral separates into a constant term and a term involving dn^2 . Evaluating each contribution yields

$$\mathcal{Y}_{a_1}^{[3]} = -\frac{2i\lambda_3^2 K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}} + \frac{2i(\lambda_1^2 - \lambda_3^2)}{\sqrt{\lambda_1^2 - \lambda_3^2}} E(u, k_2^{(1)}) \Big|_{u=K_2}^{u=0} = -\frac{2i(\lambda_3^2 K_2^{(1)} + (\lambda_1^2 - \lambda_3^2) E_2^{(1)})}{\sqrt{\lambda_1^2 - \lambda_3^2}}.$$

By symmetry of the a -cycles,

$$(10.44) \quad \mathcal{Y}_{a_2}^{[3]} = \mathcal{Y}_{a_1}^{[3]}.$$

We next turn to the b -cycle integrals. Starting from

$$(10.45) \quad \mathcal{Y}_{b_2}^{[3]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^3 d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-K_2 + iK_2'}^{-K_2} \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu - iK_2', k_2^{(1)}) \right) d\nu,$$

we employ the standard shift identities

$$(10.46) \quad \operatorname{dn}(z - K, k) = \frac{k'}{\operatorname{dn}(z, k)}, \quad \operatorname{cn}(z - K', k') = \frac{k \operatorname{sn}(z, k')}{\operatorname{dn}(z, k')}.$$

From these formulas one obtains

$$\begin{aligned}
 \operatorname{dn}(iu - K_2 - iK'_2, k_2^{(1)}) &= \frac{k_2^{(1)'}}{\operatorname{dn}(i(u - K'_2), k_2^{(1)})} \\
 &= k_2^{(1)' } \frac{\operatorname{cn}(u - K'_2, k_2^{(1)'})}{\operatorname{dn}(u - K'_2, k_2^{(1)'})} = k_2^{(1)' } \operatorname{sn}(u, k_2^{(1)'}).
 \end{aligned}
 \tag{10.47}$$

Setting $v = iu - K_2$, the integral becomes

$$\begin{aligned}
 \mathcal{Y}_{b_2}^{[3]} &= \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K'_2}^0 \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) (k_2^{(1)'})^2 \operatorname{sn}^2(u, k_2^{(1)'}) \right) i du \\
 &= \frac{2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_2} \left(\lambda_1^2 - (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(u, k_2^{(1)'}) \right) du = \frac{2\lambda_1^2 K'_2 - 2(\lambda_1^2 - \lambda_3^2) E'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}}.
 \end{aligned}
 \tag{10.48}$$

Again by symmetry of the b -cycles,

$$\mathcal{Y}_{b_1}^{[3]} = \mathcal{Y}_{b_2}^{[3]}.
 \tag{10.49}$$

Week 11: Algebraic-Geometric Solutions for the mKdV Equation (11)

2025-12-02

Objectives: Focus on Case 1. Determine the parameters $C_{u_0}^{(1)}$ and $D^{(1)}$.

We set

$$(11.1) \quad D = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}), \quad \Delta = \mathcal{A}_{P_0}(\infty^+) - \mathcal{A}_{P_0}(\infty^-) = \mathcal{A}_{\infty^-}(\infty^+), \quad \mu_0 = \mu(x_0, t_0).$$

The Abel images of the divisors D_μ and D_{μ^*} are related by

$$(11.2) \quad \alpha_{P_0}(D_{\mu^*}) = \alpha_{P_0}(D_\mu) + \Delta.$$

Moreover, the evolution of $\alpha_{P_0}(D_\mu)$ along the flow is given by

$$(11.3) \quad \alpha_{P_0}(D_\mu) = \alpha_{P_0}(D_{\mu_0}) - i\widehat{U}(x - x_0) - i\widehat{V}(t - t_0) = \alpha_{P_0}(D_{\mu_0}) - i\widehat{U}(x + vt),$$

where we have fixed $(x_0, t_0) = (0, 0)$ and used the relation

$$\widehat{V} = v\widehat{U}, \quad v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2).$$

Combining these identities with the definition of D yields the following shifted forms:

$$(11.4) \quad D + \Delta = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0}),$$

$$(11.5) \quad D + \Delta + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu),$$

$$(11.6) \quad D + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}),$$

and

$$(11.7) \quad D - \Delta + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}).$$

We now introduce the constant

$$(11.8) \quad C_{u_0} = \frac{\Theta(D)}{\Phi_{1,0}^+ \Theta(D + \Delta)}.$$

With this notation, the representation (4.10) can be rewritten as

$$\begin{aligned}
 u(x, t) &= \frac{\Theta(D) \Theta(D + \Delta + i\widehat{U}(x + vt))}{\Phi_{1,0}^+ \Theta(D + \Delta) \Theta(D + i\widehat{U}(x + vt))} e^{2i\omega_2 x + 2i\omega_3 t} \\
 &= C_{u_0} \frac{\Theta(D + \Delta + i\widehat{U}(x + vt))}{\Theta(D + i\widehat{U}(x + vt))} e^{2i\omega_2 x + 2i\omega_3 t}.
 \end{aligned}
 \tag{11.9}$$

Equivalently, starting from (4.12), one obtains

$$(11.10) \quad u(x, t) = \frac{-4 \Theta(D - \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

For Case 1.

Since

$$(11.11) \quad \widehat{U} = (U_1, U_2) = (2d_{11}, 2d_{21}), \quad d_{21} = d_{11} = -\frac{\pi \sqrt{\lambda_1^2 - \lambda_3^2}}{2K_2^{(1)}} \in \mathbb{R}^-,$$

it follows that $U_1 = U_2$. Consequently,

$$(11.12) \quad \omega_2 = -\frac{i}{4\pi} \sum_{j=1}^2 U_j \mathcal{Y}_{a_j}^{[2]} = -\frac{i}{4\pi} (U_1 \mathcal{Y}_{a_1}^{[2]} + U_2 \mathcal{Y}_{a_2}^{[2]}) = 0,$$

where we have used $\mathcal{Y}_{a_1}^{[2]} = -\mathcal{Y}_{a_2}^{[2]}$. Hence

$$(11.13) \quad \omega_3 = v \omega_2 = 0.$$

Moreover,

$$(11.14) \quad \Delta = \Delta^{(1)} = \begin{pmatrix} \pi i + \nu^{(1)} \\ \pi i - \nu^{(1)} \end{pmatrix}, \quad \nu^{(1)} = \frac{\pi i}{K_1^{(1)}} F\left(\frac{\lambda_2}{\lambda_3}, k_1^{(1)}\right) \in i\mathbb{R}.$$

From this explicit form one readily finds

$$(11.15) \quad \Delta^* = -\Delta, \quad \sigma_1 \Delta = \begin{pmatrix} \pi i - \nu^{(1)} \\ \pi i + \nu^{(1)} \end{pmatrix} = \begin{pmatrix} 2\pi i \\ 2\pi i \end{pmatrix} - \Delta, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Using (9.16) and (9.17), the period matrix satisfies

$$(11.16) \quad \sigma_1 B^{(1)} \sigma_1 = B^{(1)}, \quad \sigma_1^\top = \sigma_1, \quad \sigma_1^2 = \mathbb{I}.$$

Letting $m = \sigma_1 n$, we compute

$$(11.17) \quad \begin{aligned} \Theta(B^{(1)}, \sigma_1 z) &= \sum_{n \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} n^\top B^{(1)} n + n^\top \sigma_1 z\right\} \\ &= \sum_{n \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} (\sigma_1 n)^\top \sigma_1 B^{(1)} \sigma_1 n + (\sigma_1 n)^\top z\right\} \\ &= \sum_{m \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} m^\top B^{(1)} m + m^\top z\right\} = \Theta(B^{(1)}, z). \end{aligned}$$

Using the identities

$$\Theta^*(z) = \Theta(z^*), \quad \Theta(\sigma_1 z) = \Theta(z), \quad \Theta(-z) = \Theta(z),$$

and the representation

$$(11.18) \quad u(x, t) = C_{u_0} \frac{\Theta(D + \Delta + i\widehat{U}(x + vt))}{\Theta(D + i\widehat{U}(x + vt))},$$

we obtain

$$(11.19) \quad \begin{aligned} u^*(x, t) &= C_{u_0}^* \frac{\Theta(D^* + \Delta^* - i\widehat{U}(x + vt))}{\Theta(D^* - i\widehat{U}(x + vt))} \\ &= C_{u_0}^* \frac{\Theta(\sigma_1 D^* + \sigma_1 \Delta^* - i\widehat{U}(x + vt))}{\Theta(\sigma_1 D^* - i\widehat{U}(x + vt))} \\ &= C_{u_0}^* \frac{\Theta(-\sigma_1 D^* - \Delta + i\widehat{U}(x + vt))}{\Theta(-\sigma_1 D^* + i\widehat{U}(x + vt))}. \end{aligned}$$

Requiring $u(x, t) \in \mathbb{R}$ and comparing (11.10) with (11.19), we obtain the constraint

$$(11.20) \quad D = -\sigma_1 D^* + 2\pi i n + B^{(1)} m, \quad D = \begin{pmatrix} D_1 \\ D_2 \end{pmatrix}, \quad n, m \in \mathbb{Z}^2.$$

Equivalently, this condition can be written componentwise as

$$(11.21) \quad \begin{aligned} D_1 + D_2^* &= 2\pi i n_1 + B_{11} m_1 + B_{12} m_2, \\ D_1^* + D_2 &= 2\pi i n_2 + B_{21} m_1 + B_{22} m_2. \end{aligned}$$

Using the symmetry properties of the period matrix,

$$(11.22) \quad B_{11} = B_{22}, \quad B_{21} = B_{12}, \quad B_{ij} \in \mathbb{R},$$

and noting that $B_{11} \neq B_{12}$, we conclude that

$$(11.23) \quad n_1 = -n_2, \quad m_1 = m_2.$$

Substituting these relations back into (11.21) yields

$$(11.24) \quad D_1 + D_2^* = 2\pi i n_1 + (B_{11} + B_{12}) m_1.$$

On the other hand, rewriting $u(x, t)$ in (11.10) in the form

$$(11.25) \quad u(x, t) = \frac{-4\Theta(\sigma_1 D - \sigma_1 \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))} = \frac{-4\Theta(\sigma_1 D + \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))},$$

and comparing with (11.18), we impose the condition

$$(11.26) \quad D = \sigma_1 D + 2\pi i n', \quad n' \in \mathbb{Z}^2.$$

This is equivalent to the pair of relations

$$(11.27) \quad D_1 - D_2 = 2\pi i n'_1, \quad D_2 - D_1 = 2\pi i n'_2,$$

and hence

$$(11.28) \quad n'_2 = -n'_1, \quad D_1 - D_2 = 2\pi i n'_1.$$

Under the condition (11.26), comparison of (11.10) with (11.18) yields

$$(11.29) \quad C_{u_0}^2 = \frac{-4\Theta(D - \Delta + i\widehat{U}(x + vt))}{\omega_1^2 \Theta(D + \Delta + i\widehat{U}(x + vt))} = \frac{-4\Theta(\sigma_1 D + \Delta + i\widehat{U}(x + vt))}{\omega_1^2 \Theta(D + \Delta + i\widehat{U}(x + vt))} = \frac{-4}{\omega_1^2},$$

where the σ_1 -symmetry of the theta function has been used. Consequently,

$$(11.30) \quad C_{u_0} = \frac{2i}{\omega_1}.$$

Combining (11.24) with (11.28), we obtain the constraints

$$(11.31) \quad D_1 - D_2 = 2\pi i n'_1, \quad D_1 + D_2^* = 2\pi i n_1 + (B_{11} + B_{12}) m_1.$$

These imply

$$(11.32) \quad \operatorname{Re} D_1 = \operatorname{Re} D_2 = \frac{1}{2} m_1 (B_{11} + B_{12}), \quad \operatorname{Im} D_1 = \operatorname{Im} D_2 + 2\pi n_1.$$

Equivalently, the vector D admits the parameterization

$$(11.33) \quad \begin{pmatrix} D_1 \\ D_2 \end{pmatrix} = \frac{m_1}{2} B^{(1)} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 1 \end{pmatrix} \alpha + \begin{pmatrix} 2\pi i n_1 \\ 0 \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

By (11.20), we have

$$(11.34) \quad -\sigma_1 D^* = D - 2\pi i n - B^{(1)} m.$$

Under this condition, the complex conjugate of $u(x, t)$ in (11.19) can be written as

$$\begin{aligned}
 u^*(x, t) &= C_{u_0}^* \frac{\Theta(D - 2\pi i n - B^{(1)}m - \Delta + i\hat{U}(x + vt))}{\Theta(D - 2\pi i n - B^{(1)}m + i\hat{U}(x + vt))} \\
 (11.35) \quad &= C_{u_0}^* \frac{\Theta(D - \Delta + i\hat{U}(x + vt)) \exp\left\{\langle D - \Delta + i\hat{U}(x + vt), m \rangle + \frac{1}{2}(B_{11}m_1 + B_{22}m_2)\right\}}{\Theta(D + i\hat{U}(x + vt)) \exp\left\{\langle D + i\hat{U}(x + vt), m \rangle + \frac{1}{2}(B_{11}m_1 + B_{22}m_2)\right\}} \\
 &= C_{u_0}^* \frac{\Theta(D - \Delta + i\hat{U}(x + vt)) \exp\{-\langle \Delta, m \rangle\}}{\Theta(D + i\hat{U}(x + vt))}.
 \end{aligned}$$

Here $m_2 = m_1$, and therefore

$$\begin{aligned}
 (11.36) \quad \exp\{-\langle \Delta, m \rangle\} &= \exp\left\{-(\pi i + v^{(1)})m_1 - (\pi i - v^{(1)})m_1\right\} \\
 &= \exp(-2\pi i m_1) = 1.
 \end{aligned}$$

It then follows from (11.10) and (11.35) that

$$(11.37) \quad |C_{u_0}|^2 = \frac{-4}{\omega_1^2}.$$

Week 12: Algebro-Geometric Solutions for the mKdV Equation (12)

2025-12-09

Objectives: Focus on Case 1. Introduce the effective integration method.

Recall that

$$(12.1) \quad L(\lambda; x, t) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

whose entries are given explicitly by

$$\begin{aligned}
 (12.2) \quad L_{11}(\lambda) &= -i(\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2)), \quad L_{22}(\lambda) = -L_{11}(\lambda), \\
 L_{12}(\lambda) &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\
 L_{21}(\lambda) &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*).
 \end{aligned}$$

The effective integration method amounts to exploiting the fact that the x -part of (12.3) is an ordinary differential equation,

$$u_{xxx} + 6u^2u_x - 4\alpha_1u_x = 0,$$

which is encoded in the equations

$$(12.3) \quad L_x = [U(\lambda, Q), L], \quad L_t = [V(\lambda, Q), L].$$

We compute the determinant of L . Using the structure of the matrix,

$$\begin{aligned}
 (12.4) \quad \det L &= -L_{11}^2 - L_{12}L_{21} \\
 &= \lambda^2(\lambda^2 + \alpha_1 - \frac{1}{2}u^2)^2 + u^2\left[\left(\lambda^2 - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right)^2 + \frac{u_x^2}{4u^2}\lambda^2\right].
 \end{aligned}$$

A direct expansion yields the polynomial form

$$\begin{aligned}
 (12.5) \quad \det L &= \lambda^6 + 2\alpha_1\lambda^4 + \left(\frac{1}{4}(u_x^2 - 2uu_{xx} - 3u^4) + \alpha_1u^2 + \alpha_1^2\right)\lambda^2 + \left(\frac{u_{xx}}{4} - \alpha_1u + \frac{u^3}{2}\right)^2 \\
 &= \lambda^6 + 2\alpha_1\lambda^4 + s_2\lambda^2 + s_0.
 \end{aligned}$$

Identifying coefficients in (12.5) gives, at order $O(1)$,

$$(12.6) \quad u_{xx} - 4\alpha_1 u + 2u^3 = 4\sqrt{s_0},$$

and, at order $O(\lambda^2)$,

$$(12.7) \quad u_x^2 - 2uu_{xx} - 3u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2.$$

Substituting (12.6) into (12.7) eliminates u_{xx} and yields

$$(12.8) \quad u_x^2 - 2u(4\sqrt{s_0} + 4\alpha_1 u - 2u^3) - 3u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2,$$

which simplifies to

$$(12.9) \quad u_x^2 - 8u\sqrt{s_0} - 8\alpha_1 u^2 + u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2.$$

Equivalently,

$$(12.10) \quad \begin{aligned} u_x^2 &= -(u^4 - 4\alpha_1 u^2 - 8u\sqrt{s_0} + 4\alpha_1^2 - 4s_2) \\ &= -R(u) = -\prod_{i=1}^4 (u - u_i), \end{aligned}$$

from which it follows immediately that

$$(12.11) \quad \sum_{i=1}^4 u_i = 0.$$

Multiplying (12.6) by u_x yields

$$(12.12) \quad u_x u_{xx} = 4\alpha_1 u u_x + 2u^3 u_x = 4\sqrt{s_0} u_x.$$

Integrating once with respect to x , we obtain

$$(12.13) \quad u_x^2 = -(-4\alpha_1 u^2 + u^4 - 8\sqrt{s_0} u) + c_0, \quad c_0 = 4s_2 - 4\alpha_1^2,$$

which is precisely equivalent to (12.10). From a different perspective, the conserved energy is

$$(12.14) \quad E = 4s_2 - 4\alpha_1^2 = u_x^2 + (u^4 - 4\alpha_1 u^2 - 8\sqrt{s_0} u).$$

Since $u_x^2 = -R(u)$, we may write equivalently

$$(12.15) \quad \frac{du}{dx} = \pm \sqrt{-R(u)}.$$

Separating variables gives

$$(12.16) \quad \int \frac{du}{\sqrt{-R(u)}} = \pm x + C.$$

Note that u and u_x are real-valued.

Case I. $s_0 \neq 0$, $u_i \in \mathbb{R}$, $u_1 > u_2 > u_3 > u_4$. In this case there are two types of real solutions, corresponding to the admissible intervals

$$(12.17) \quad u_2 \leq u^{[1]} \leq u_1, \quad u_4 \leq u^{[2]} \leq u_3.$$

Case II. $s_0 \neq 0$, $u_{1,3} \in \mathbb{R}$, $u_1 > u_3$, and $u_{2,4} \in \mathbb{C} \setminus \mathbb{R}$. Then the real solution satisfies

$$(12.18) \quad u_3 \leq u^{[3]} \leq u_1.$$

Case III. $s_0 = 0$, $u_4 = -u_1$, $u_3 = -u_2$ (genus one).

We now describe the correspondence between the branch points λ_i and the roots u_i . If $R(u_i) = 0$, then $\mu_1^* = \mu_1$ and $\mu_2^* = \mu_2$, and the determinant of L takes the form

$$\begin{aligned}
 \det L &= \lambda^2 \left(\lambda^2 + \alpha_1 - \frac{u_i^2}{2} \right)^2 + u_i^2 (\lambda - \mu_1)^2 (\lambda - \mu_2)^2 \\
 (12.19) \quad &= \left(\lambda^3 + \lambda \left(\alpha_1 - \frac{u_i^2}{2} \right) + i u_i (\lambda - \mu_1) (\lambda - \mu_2) \right) \\
 &\quad \times \left(\lambda^3 + \lambda \left(\alpha_1 - \frac{u_i^2}{2} \right) - i u_i (\lambda - \mu_1) (\lambda - \mu_2) \right).
 \end{aligned}$$

We wish to decompose $\det L = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)$ into two cubic polynomials of the above form. This requires that the two factors have opposite coefficients in the quadratic and constant terms. In the Case 1, all λ_j lie on the imaginary axis, $\lambda_j \in i\mathbb{R}$. In the Case 2, one has $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$, $\lambda_2 \in i\mathbb{R}$, and $\lambda_3 = -\lambda_1^*$.

For distinct $\lambda_i, \lambda_j, \lambda_k$, the identities

$$\begin{aligned}
 (12.20) \quad &(\lambda^2 - \lambda_i^2)(\lambda - \lambda_j) = \lambda^3 - \lambda_j \lambda^2 - \lambda_i^2 \lambda + \lambda_i^2 \lambda_j, \\
 &(\lambda^2 - \lambda_k^2)(\lambda + \lambda_j) = \lambda^3 + \lambda_j \lambda^2 - \lambda_k^2 \lambda - \lambda_k^2 \lambda_j
 \end{aligned}$$

show that the constant terms cannot be matched in this manner.

Instead, consider $S_i, S_j, S_k \in \{\pm 1\}$ and distinct $\lambda_i, \lambda_j, \lambda_k$. Then

$$(12.21) \quad (\lambda - S_i \lambda_i)(\lambda - S_j \lambda_j)(\lambda - S_k \lambda_k) = \lambda^3 - (S_i \lambda_i + S_j \lambda_j + S_k \lambda_k) \lambda^2 + \cdots - S_i S_j S_k \lambda_i \lambda_j \lambda_k,$$

and

$$(12.22) \quad (\lambda + S_i \lambda_i)(\lambda + S_j \lambda_j)(\lambda + S_k \lambda_k) = \lambda^3 + (S_i \lambda_i + S_j \lambda_j + S_k \lambda_k) \lambda^2 + \cdots + S_i S_j S_k \lambda_i \lambda_j \lambda_k.$$

Comparing coefficients with (12.19), we obtain the relations

$$\begin{aligned}
 (12.23) \quad &i u_1 = \lambda_1 + \lambda_2 + \lambda_3, \quad i u_2 = -(\lambda_1 + \lambda_2 - \lambda_3), \\
 &i u_3 = -(\lambda_1 - \lambda_2 + \lambda_3), \quad i u_4 = -(-\lambda_1 + \lambda_2 + \lambda_3).
 \end{aligned}$$

Supplementary computations:

Set $\sqrt{s_0} = i \lambda_1 \lambda_2 \lambda_3$. With this choice, the quartic equation admits four roots given explicitly by

$$u_1 = -i(\lambda_1 + \lambda_2 + \lambda_3), \quad u_2 = i(\lambda_1 + \lambda_2 - \lambda_3), \quad u_3 = i(\lambda_1 - \lambda_2 + \lambda_3), \quad u_4 = i(-\lambda_1 + \lambda_2 + \lambda_3).$$

Pairing the factors gives

$$(u - u_1)(u - u_2) = (u + i(\lambda_1 + \lambda_2 + \lambda_3))(u - i(\lambda_1 + \lambda_2 - \lambda_3)) = u^2 + 2i\lambda_3 u + (\lambda_1 + \lambda_2)^2 - \lambda_3^2.$$

$$(u - u_3)(u - u_4) = (u - i(\lambda_1 - \lambda_2 + \lambda_3))(u - i(-\lambda_1 + \lambda_2 + \lambda_3)) = u^2 - 2i\lambda_3 u + (\lambda_1 - \lambda_2)^2 - \lambda_3^2.$$

Hence

$$\begin{aligned}
 \prod_{i=1}^4 (u - u_i) &= (u^2 + 2i\lambda_3 u + (\lambda_1 + \lambda_2)^2 - \lambda_3^2)(u^2 - 2i\lambda_3 u + (\lambda_1 - \lambda_2)^2 - \lambda_3^2) \\
 &= u^4 + 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)u^2 - 8i\lambda_1 \lambda_2 \lambda_3 u + ((\lambda_1^2 + \lambda_2^2 + \lambda_3^2)^2 - 4(\lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2)).
 \end{aligned}$$

Using $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = -2\alpha_1$, $\lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2 = s_2$, and $\sqrt{s_0} = i \lambda_1 \lambda_2 \lambda_3$, we obtain

$$\prod_{i=1}^4 (u - u_i) = u^4 - 4\alpha_1 u^2 - 8\sqrt{s_0} u + 4\alpha_1^2 - 4s_2 = R(u),$$

so each u_i is a root of $R(u) = 0$.

Objectives: Derive explicit expressions for u by integrating the first-order equation $u_x^2 = -R(u)$ and reducing the resulting integrals to standard elliptic forms.

We begin with

$$(13.1) \quad u_x^2 = -R(u) = -\prod_{i=1}^4 (u - u_i).$$

We solve this equation by direct integration, distinguishing cases according to the location of the roots.

Case 1. All roots are real, $u_i \in \mathbb{R}$ (equivalently $\lambda_i \in i\mathbb{R}$), ordered as

$$u_1 > u_2 > u_3 > u_4.$$

We consider the initial value problem on the interval $u_4 \leq u \leq u_3$,

$$(13.2) \quad \begin{cases} \frac{du}{dx} = \pm \sqrt{-R(u)}, \\ u(0) = u_3. \end{cases}$$

Setting

$$y = u_x, \quad z = u,$$

the corresponding algebraic curve takes the form

$$(13.3) \quad y^2 = -\prod_{i=1}^4 (z - u_i).$$

Integrating (13.2) yields

$$(13.4) \quad \int_{u_3}^u \frac{dv}{\sqrt{-R(v)}} = \pm \int_0^x dy = \pm x.$$

We recall the incomplete elliptic integral of the first kind,

$$(13.5) \quad v = F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad z = \text{sn}(v, k).$$

Differentiating with respect to v gives

$$1 = \frac{1}{\sqrt{(1-z^2)(1-k^2z^2)}} \frac{dz}{dv},$$

and hence

$$(13.6) \quad \frac{dz}{dv} = \sqrt{(1-z^2)(1-k^2z^2)} = \text{cn}(v, k) \text{dn}(v, k),$$

where

$$\text{cn}(v, k) = \sqrt{1-z^2}, \quad \text{dn}(v, k) = \sqrt{1-k^2z^2}.$$

To reduce the integral (13.4) to the standard form, we introduce a fractional linear transformation $u \mapsto t$, followed by the substitution $t = z^2$, such that

$$u_3 \leftrightarrow 0, \quad u_4 \leftrightarrow 1, \quad u_2 \leftrightarrow \infty.$$

An explicit choice is

$$(13.7) \quad t = \frac{u - u_3}{u - u_2} \frac{u_4 - u_2}{u_4 - u_3}.$$

Solving for u gives

$$(13.8) \quad u = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{u_4 - u_2 + (u_3 - u_4)t}, \quad t = z^2.$$

From (13.8), straightforward algebra yields

$$(13.9) \quad u - u_1 = \frac{(u_3 - u_1)(u_4 - u_2) + (u_2 - u_1)(u_3 - u_4)t}{u_4 - u_2 + (u_3 - u_4)t},$$

$$(13.10) \quad u - u_3 = \frac{(u_2 - u_3)(u_3 - u_4)t}{u_4 - u_2 + (u_3 - u_4)t},$$

and

$$(13.11) \quad u - u_4 = \frac{(u_2 - u_4)(u_4 - u_3)(1 - t)}{u_4 - u_2 + (u_3 - u_4)t}.$$

Combining these expressions, we find

$$(13.12) \quad \prod_{i=1}^4 (u - u_i) = \frac{-(u_2 - u_3)^2(u_2 - u_4)^2(u_3 - u_4)^2 t(1 - t) \left((u_1 - u_3)(u_2 - u_4) + (u_2 - u_1)(u_3 - u_4)t \right)}{(u_4 - u_2 + (u_3 - u_4)t)^4}.$$

Consequently,

$$(13.13) \quad \sqrt{-R(u)} = \frac{(u_2 - u_3)(u_2 - u_4)(u_3 - u_4)}{(u_4 - u_2 + (u_3 - u_4)t)^2} \sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{t(1 - t)(1 - k^2 t)},$$

where the elliptic modulus is given by

$$(13.14) \quad k^2 = \frac{(u_1 - u_2)(u_3 - u_4)}{(u_1 - u_3)(u_2 - u_4)}.$$

Moreover, differentiating (13.8) with respect to t yields

$$(13.15) \quad du = \frac{(u_2 - u_4)(u_2 - u_3)(u_4 - u_3)}{(u_4 - u_2 + (u_3 - u_4)t)^2} dt.$$

Substituting this expression together with (13.13) and setting $t = z^2$, we obtain

$$(13.16) \quad \frac{du}{\sqrt{-R(u)}} = \frac{-dt}{\sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{t(1 - t)(1 - k^2 t)}}$$

$$(13.17) \quad = \frac{-2dz}{\sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{(1 - z^2)(1 - k^2 z^2)}}, \quad z \in (0, 1).$$

Consequently, for $u \in [u_4, u_3]$,

$$(13.18) \quad \int_{u_3}^u \frac{dv}{\sqrt{-R(v)}} = -\frac{2}{\sqrt{(u_1 - u_3)(u_2 - u_4)}} \int_0^z \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}$$

$$(13.19) \quad = -\frac{2}{\sqrt{(u_1 - u_3)(u_2 - u_4)}} F(z, k) = \pm x.$$

Inverting this relation gives

$$(13.20) \quad z^2 = \text{sn}^2(\alpha x, k), \quad \alpha^2 = \frac{(u_1 - u_3)(u_2 - u_4)}{4}.$$

Substituting back into (13.8) yields the explicit solution

$$(13.21) \quad u(x) = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{(u_4 - u_2) + (u_3 - u_4) \text{sn}^2(\alpha x, k)}, \quad x \in \mathbb{R}, \quad u \in [u_4, u_3].$$

Case 2. Assume

$$u_1 > u_3 \in \mathbb{R}, \quad u_2 = u_4^* \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R}).$$

Then

$$(13.22) \quad \begin{aligned} -R(u) &= -(u - u_3)(u - u_1)(u - u_2)(u - u_2^*) \\ &= -(u - u_3)(u - u_1)((u - u_{2R})^2 + u_{2I}^2), \end{aligned}$$

where $u_2 = u_{2R} + iu_{2I} = a + ib$.

Using the identity $\text{cn}'(x, k) = -\text{sn}(x, k) \text{dn}(x, k)$, we set

$$z = \text{cn}(x, k).$$

Then

$$(13.23) \quad \left(\frac{dz}{dx}\right)^2 = \text{sn}^2(x, k) \text{dn}^2(x, k) = (1 - z^2)(k'^2 + k^2 z^2),$$

where $k' = \sqrt{1 - k^2}$. It follows that

$$(13.24) \quad \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}} = -dx.$$

Integrating from $x = 0$ (so that $z = 1$) to x gives

$$(13.25) \quad -\int_1^z \frac{dt}{\sqrt{(1 - t^2)(k'^2 + k^2 t^2)}} = \int_0^x dy = x.$$

To relate z to u , we introduce a Möbius transformation mapping

$$u_1 \leftrightarrow 1, \quad u_3 \leftrightarrow -1.$$

Let

$$(13.26) \quad \delta \frac{u - u_1}{u - u_3} = \frac{z - 1}{z + 1}.$$

Solving for u gives

$$(13.27) \quad u = \frac{u_1 - \frac{z - 1}{\delta(z + 1)} u_3}{1 - \frac{z - 1}{\delta(z + 1)}} = u_1 + \frac{(u_1 - u_3) \frac{z - 1}{\delta(z + 1)}}{1 - \frac{z - 1}{\delta(z + 1)}}.$$

Equivalently, this can be written as

$$(13.28) \quad u = u_1 + \frac{(u_3 - u_1)(1 - z)}{(1 - z) + \delta(1 + z)} = u_1 + \frac{(u_3 - u_1)(1 - z)}{1 + \delta + (\delta - 1)z}.$$

Supplementary computations:

To simplify the computations, we employ the complex projective line notation on $\mathbb{CP}^1$. We write a Möbius transformation as

$$\frac{az + bw}{cz + dw} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}, \quad \begin{pmatrix} z \\ w \end{pmatrix} = (z : w) = \frac{z}{w} \in \mathbb{CP}^1.$$

For any nonzero scalar f , this representation is homogeneous:

$$\begin{pmatrix} fa & fb \\ fc & fd \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = f \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}.$$

Moreover, if two matrices share the same second row, then

$$\begin{pmatrix} a_1 & b_1 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}.$$

From (13.26) we have the matrix identity

$$(13.29) \quad \begin{pmatrix} 1 & -u_1 \\ 1 & -u_3 \end{pmatrix} \begin{pmatrix} u \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ \delta & \delta \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Multiplying by the inverse of the left matrix yields

$$(13.30) \quad \begin{pmatrix} u \\ 1 \end{pmatrix} = \frac{1}{u_1 - u_3} \begin{pmatrix} -u_3 & u_1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ \delta & \delta \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} = \begin{pmatrix} \delta u_1 - u_3 & \delta u_1 + u_3 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Equivalently,

$$(13.31) \quad u = \frac{(\delta u_1 - u_3)z + \delta u_1 + u_3}{(\delta - 1)z + \delta + 1} = \frac{\delta u_1(z + 1) + u_3(1 - z)}{\delta(z + 1) + 1 - z}.$$

Subtracting $\begin{pmatrix} u_2 \\ 1 \end{pmatrix}$ from (13.30) gives

$$(13.32) \quad \begin{pmatrix} u \\ 1 \end{pmatrix} - \begin{pmatrix} u_2 \\ 1 \end{pmatrix} = \begin{pmatrix} \delta u_1 - u_3 & \delta u_1 + u_3 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} - \begin{pmatrix} u_2(\delta - 1) & u_2(\delta + 1) \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} \\ = \begin{pmatrix} \delta(u_1 - u_2) + u_2 - u_3 & \delta(u_1 - u_2) + u_3 - u_2 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Recall that multiplication in $\mathbb{CP}^1$ is represented componentwise:

$$\begin{pmatrix} z_1 \\ w_1 \end{pmatrix} \cdot \begin{pmatrix} z_2 \\ w_2 \end{pmatrix} = \begin{pmatrix} z_1 z_2 \\ w_1 w_2 \end{pmatrix} = \frac{z_1 z_2}{w_1 w_2}.$$

Similarly,

$$(13.33) \quad \begin{pmatrix} u - u_2^* \\ 1 \end{pmatrix} = \begin{pmatrix} \delta(u_1 - u_2^*) + u_2^* - u_3 & \delta(u_1 - u_2^*) + u_3 - u_2^* \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

It follows that the denominator of $(u - u_2)(u - u_2^*)$ is

$$(13.34) \quad ((\delta - 1)z + \delta + 1)^2,$$

while the numerator is

$$(13.35) \quad (z - 1) \begin{pmatrix} \delta(u_1 - u_2) + u_2 - u_3 \\ \delta(u_1 - u_2) + u_3 - u_2 \end{pmatrix} \begin{pmatrix} \delta(u_1 - u_2^*) + u_2^* - u_3 & \delta(u_1 - u_2^*) + u_3 - u_2^* \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Requiring the coefficient of the linear term in z to vanish leads to

$$(13.36) \quad \operatorname{Re}((\delta(u_1 - u_2) + u_2 - u_3)(\delta(u_1 - u_2^*) + u_3 - u_2^*)) = 0.$$

Equivalently,

$$(13.37) \quad \delta^2 |u_1 - u_2|^2 + \operatorname{Re}((u_2 - u_3)(u_1 - u_2^*) - (u_2^* - u_3)(u_1 - u_2)) \delta - |u_2 - u_3|^2 = 0.$$

Using

$$\operatorname{Re}((u_2 - u_3)(u_1 - u_2^*) - (u_2^* - u_3)(u_1 - u_2)) = \operatorname{Re}(2i \operatorname{Im}(u_2 - u_3)(u_1 - u_2)) = 0,$$

we obtain

$$(13.38) \quad \delta^2 = \frac{|u_2 - u_3|^2}{|u_1 - u_2|^2}, \quad \delta = \frac{|u_2 - u_3|}{|u_1 - u_2|}.$$

Hence

$$(13.39) \quad (u - u_2)(u - u_2^*) = \frac{|\delta(u_1 - u_2) + u_2 - u_3|^2 z^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2}{((\delta - 1)z + \delta + 1)^2}.$$

Moreover,

$$\begin{pmatrix} u - u_1 \\ 1 \end{pmatrix} = \begin{pmatrix} u_1 - u_3 & u_3 - u_1 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}, \quad \begin{pmatrix} u - u_3 \\ 1 \end{pmatrix} = \begin{pmatrix} \delta(u_1 - u_3) & \delta(u_1 - u_3) \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix},$$

so that

$$(13.40) \quad u - u_1 = \frac{(u_1 - u_3)(z - 1)}{(\delta - 1)z + \delta + 1}, \quad u - u_3 = \frac{\delta(u_1 - u_3)(z + 1)}{(\delta - 1)z + \delta + 1}.$$

Substituting (13.39) and (13.40) into (13.22), we obtain

$$(13.41) \quad -R(u) = \frac{\delta(u_1 - u_3)^2(1 - z)(1 + z)}{((\delta - 1)z + \delta + 1)^4} \left(|\delta(u_1 - u_2) + u_2 - u_3|^2 z^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2 \right).$$

Using the identity, valid for all $z_1, z_2 \in \mathbb{C}$,

$$(13.42) \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2,$$

we find

$$(13.43) \quad \begin{aligned} & |\delta(u_1 - u_2) + u_2 - u_3|^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2 \\ &= 2\delta^2|u_1 - u_2|^2 + 2|u_2 - u_3|^2 = 4|u_2 - u_3|^2, \end{aligned}$$

where we used $\delta^2|u_1 - u_2|^2 = |u_2 - u_3|^2$.

Consequently,

$$(13.44) \quad \sqrt{-R(u)} = \frac{2\sqrt{\delta}(u_1 - u_3)|u_2 - u_3|}{((\delta - 1)z + \delta + 1)^2} \sqrt{(1 - z^2)(k'^2 + k^2 z^2)},$$

where

$$(13.45) \quad k = \frac{|\delta(u_1 - u_2) + u_2 - u_3|}{2|u_2 - u_3|}, \quad k' = \sqrt{1 - k^2}.$$

Differentiating (13.31) with respect to z gives

$$(13.46) \quad \begin{aligned} du &= \frac{(\delta u_1 - u_3)((\delta - 1)z + \delta + 1) - ((\delta u_1 - u_3)z + \delta u_1 + u_3)(\delta - 1)}{((\delta - 1)z + \delta + 1)^2} dz \\ &= \frac{2\delta(u_1 - u_3)}{((\delta - 1)z + \delta + 1)^2} dz. \end{aligned}$$

Combining this with (13.44), we obtain

$$(13.47) \quad \frac{du}{\sqrt{-R(u)}} = \frac{\sqrt{\delta}}{|u_2 - u_3|} \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}} = \frac{1}{\alpha} \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}},$$

where

$$(13.48) \quad \alpha = \frac{|u_2 - u_3|}{\sqrt{\delta}} = \sqrt{|u_1 - u_2||u_2 - u_3|}.$$

For $u_3 \leq u \leq u_1$, integration yields

$$(13.49) \quad \int_{u_1}^u \frac{dv}{\sqrt{-R(v)}} = \frac{1}{\alpha} \int_1^z \frac{dt}{\sqrt{(1 - t^2)(k'^2 + k^2 t^2)}} = \pm x.$$

Since cn is an even function, inversion gives

$$(13.50) \quad z = \text{cn}(\alpha x, k).$$

Substituting this expression into (13.31), we arrive at the explicit solution

$$(13.51) \quad u(x) = u_1 + \frac{(u_3 - u_1)(1 - \text{cn}(\alpha x, k))}{1 + \delta + (\delta - 1)\text{cn}(\alpha x, k)}, \quad x \in \mathbb{R}, \quad u \in [u_3, u_1].$$

The parameters are given by

$$(13.52) \quad \delta = \frac{|u_2 - u_3|}{|u_1 - u_2|}, \quad \alpha^2 = |(u_1 - u_2)(u_2 - u_3)|, \quad k^2 = \frac{|\delta(u_1 - u_2) + u_2 - u_3|^2}{4|u_2 - u_3|^2}.$$

Week 14: Algebraic-Geometric Solutions for the mKdV Equation (14)

2025-12-23

Objectives: Consider Case 1 with $\lambda_j \in i\mathbb{R}$. Verify by direct computation that the two explicit expressions for the solution u coincide.

Recall from (13.21) that for $x \in \mathbb{R}$ and $u \in [u_4, u_3]$ one has

$$(14.1) \quad u(x, t) = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{(u_4 - u_2) + (u_3 - u_4) \text{sn}^2(\alpha(x + vt), k)},$$

where the parameters are given by

$$(14.2) \quad k^2 = \frac{(u_1 - u_2)(u_3 - u_4)}{(u_1 - u_3)(u_2 - u_4)}, \quad \alpha^2 = \frac{(u_1 - u_3)(u_2 - u_4)}{4}.$$

The constants u_j are expressed in terms of λ_j as

$$(14.3) \quad \begin{aligned} u_1 &= -i(\lambda_1 + \lambda_2 + \lambda_3), & u_2 &= i(\lambda_1 + \lambda_2 - \lambda_3), \\ u_3 &= i(\lambda_1 - \lambda_2 + \lambda_3), & u_4 &= i(-\lambda_1 + \lambda_2 + \lambda_3). \end{aligned}$$

On the other hand, the theta-functional representation (11.18) reads

$$(14.4) \quad u(x, t) = C_{u_0}^{(1)} \frac{\Theta(D^{(1)} + \Delta^{(1)} + iU^{(1)}(x + vt))}{\Theta(D^{(1)} + iU^{(1)}(x + vt))},$$

with

$$U^{(1)} = \kappa^{(1)} \mathbf{1}, \quad \Delta^{(1)} = (i\pi + \nu^{(1)} \sigma_3) \mathbf{1}, \quad \mathbf{1} = [1 \ 1]^\top, \quad D^{(1)} = \mathbf{0},$$

and

$$\kappa^{(1)} = \frac{\pi(\lambda_1^2 - \lambda_3^2)^{1/2}}{K_2^{(1)}}, \quad k_2^{(1)} = \frac{(\lambda_1^2 - \lambda_2^2)^{1/2}}{(\lambda_1^2 - \lambda_3^2)^{1/2}}, \quad k_1^{(1)} = k_2^{(1)} \frac{\lambda_3}{\lambda_2}, \quad \nu^{(1)} = \frac{i\pi}{K_1^{(1)}} F\left(\frac{\lambda_2}{\lambda_3}, k_1^{(1)}\right).$$

Using (7.17) and (7.24), the moduli $k_1^{(1)}$ and $k_2^{(1)}$ can be written equivalently as

$$(14.5) \quad k_1^{(1)} = \frac{\lambda_3}{\lambda_2} \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}.$$

One verifies directly that

$$k = k_2^{(1)}, \quad \alpha = \kappa^{(1)} \frac{K_2^{(1)}}{\pi}.$$

Applying the theta-function decomposition (9.22),

$$(14.6) \quad \Theta(B^{(1)}; z) = \vartheta_3(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_3(z_1 + z_2, 2\tau_2^{(1)}) + \vartheta_2(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_2(z_1 + z_2, 2\tau_2^{(1)}),$$

we first write out explicitly the numerator and denominator of (14.4). Using $D^{(1)} = \mathbf{0}$, $\Delta^{(1)} = (i\pi + \nu^{(1)} \sigma_3) \mathbf{1}$, and $U^{(1)} = \kappa^{(1)} \mathbf{1}$, we obtain

$$(14.7) \quad \begin{aligned} \Theta(D^{(1)} + \Delta^{(1)} + iU^{(1)}(x + vt)) &= \Theta((i\pi + \nu^{(1)} \sigma_3) \mathbf{1} + i\kappa^{(1)} \mathbf{1}(x + vt)) \\ &= \vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt) + 2\pi i, 2\tau_2^{(1)}) + \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt) + 2\pi i, 2\tau_2^{(1)}) \\ &= \vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}) - \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}), \end{aligned}$$

and

$$(14.8) \quad \begin{aligned} \Theta(D^{(1)} + iU^{(1)}(x + vt)) &= \Theta(i\kappa^{(1)} \mathbf{1}(x + vt)) \\ &= \vartheta_3(0, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}) + \vartheta_2(0, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}). \end{aligned}$$

Substituting (14.7) and (14.8) into (14.4) and dividing numerator and denominator by $\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})$, the ratio simplifies to

$$(14.9) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) - \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{\vartheta_3(0, 2\tau_1^{(1)}) + \vartheta_2(0, 2\tau_1^{(1)}) \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

Equivalently, factoring out the theta-constants $\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})$ and $\vartheta_2(0, 2\tau_1^{(1)})$ from the numerator and denominator, respectively, yields

$$(14.10) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} \frac{\frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})} - \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{\frac{\vartheta_3(0, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} + \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

Introduce the parameters $\tilde{v}_1$ and $\tilde{v}_2$ by

$$(14.11) \quad \tilde{v}_1 = \frac{i\pi}{K_2^{(1)}} F\left(\frac{(u_2 - u_4)^{1/2}}{(u_3 - u_4)^{1/2}}, k_2^{(1)}\right), \quad \tilde{v}_2 = \frac{i\pi}{K_2^{(1)}} F\left(\frac{(u_3(u_2 - u_4))^{1/2}}{(u_2(u_3 - u_4))^{1/2}}, k_2^{(1)}\right).$$

By (14.1), the solution can be rewritten as

$$(14.12) \quad u(x, t) = u_2 \frac{\frac{u_3(u_2 - u_4)}{u_2(u_3 - u_4)} - \text{sn}^2(\alpha(x + vt), k)}{\frac{(u_2 - u_4)}{(u_3 - u_4)} - \text{sn}^2(\alpha(x + vt), k)}.$$

Using the definitions (14.11), the constant ratios in (14.12) admit the representations

$$\frac{u_2 - u_4}{u_3 - u_4} = \text{sn}^2(\tilde{v}_1 K_2^{(1)} / (i\pi), k_2^{(1)}), \quad \frac{u_3(u_2 - u_4)}{u_2(u_3 - u_4)} = \text{sn}^2(\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}).$$

Substituting these expressions yields

$$(14.13) \quad u(x, t) = u_2 \frac{\text{sn}^2(\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}) - \text{sn}^2(\alpha(x + vt), k_2^{(1)})}{\text{sn}^2(\tilde{v}_1 K_2^{(1)} / (i\pi), k_2^{(1)}) - \text{sn}^2(\alpha(x + vt), k_2^{(1)})}.$$

Lemma 1.

$$(a) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}.$$

$$(b) \quad \frac{\vartheta_2(2\tilde{v}_1, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_1, 2\tau_2^{(1)})} = -\frac{\vartheta_3(0, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})}.$$

Proof. We begin with the three-term bilinear identities

$$(14.14) \quad \begin{aligned} 2\vartheta_3(u + v, 2\tau)\vartheta_3(u - v, 2\tau) &= \vartheta_3(u, \tau)\vartheta_3(v, \tau) + \vartheta_4(u, \tau)\vartheta_4(v, \tau), \\ 2\vartheta_2(u + v, 2\tau)\vartheta_3(u - v, 2\tau) &= \vartheta_2(u, \tau)\vartheta_2(v, \tau) - \vartheta_1(u, \tau)\vartheta_1(v, \tau). \end{aligned}$$

Setting $u = 2\tilde{v}_2$ and $v = 0$, these identities reduce to

$$(14.15) \quad \begin{aligned} 2\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})\vartheta_3(0, 2\tau_2^{(1)}) &= \vartheta_3(2\tilde{v}_2, \tau_2^{(1)})\vartheta_3(0, \tau_2^{(1)}) + \vartheta_4(2\tilde{v}_2, \tau_2^{(1)})\vartheta_4(0, \tau_2^{(1)}), \\ 2\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})\vartheta_3(0, 2\tau_2^{(1)}) &= \vartheta_2(2\tilde{v}_2, \tau_2^{(1)})\vartheta_2(0, \tau_2^{(1)}) - \vartheta_1(2\tilde{v}_2, \tau_2^{(1)})\vartheta_1(0, \tau_2^{(1)}). \end{aligned}$$

Since $\vartheta_1(0, \tau_2^{(1)}) = 0$, the last term in the second identity vanishes. Dividing the second identity by the first, we obtain

$$(14.16) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{\vartheta_2(2\tilde{v}_2, \tau_2^{(1)})\vartheta_2(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)})\vartheta_3(0, \tau_2^{(1)}) + \vartheta_4(2\tilde{v}_2, \tau_2^{(1)})\vartheta_4(0, \tau_2^{(1)})}.$$

We now express the right-hand side in terms of Jacobi elliptic functions. Using the standard relations

$$(14.17) \quad \begin{aligned} \operatorname{cn}(u, k) &= \frac{\vartheta_4(0, \tau) \vartheta_2(\pi i u / K, \tau)}{\vartheta_2(0, \tau) \vartheta_4(\pi i u / K, \tau)}, & \operatorname{dn}(u, k) &= \frac{\vartheta_4(0, \tau) \vartheta_3(\pi i u / K, \tau)}{\vartheta_3(0, \tau) \vartheta_4(\pi i u / K, \tau)}, \\ k &= \frac{\vartheta_2^2(0, \tau)}{\vartheta_3^2(0, \tau)}, & k' &= \frac{\vartheta_4^2(0, \tau)}{\vartheta_3^2(0, \tau)}, \end{aligned}$$

with $\tau = \tau_2^{(1)}$ and $u = 2\tilde{v}_2 K_2^{(1)} / (i\pi)$, we obtain

$$\frac{\vartheta_2(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_2(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_3(0, \tau_2^{(1)})} = k_2^{(1)} \frac{\operatorname{cn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})},$$

and

$$\frac{\vartheta_4(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_4(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_3(0, \tau_2^{(1)})} = \frac{k_2^{(1)'}}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}.$$

Substituting these expressions into (14.16) yields

$$(14.18) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{k_2^{(1)} \operatorname{cn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}) + k_2^{(1)'}}.$$

Applying the double-angle identities

$$(14.19) \quad \operatorname{cn}(2u) = \frac{\operatorname{cn}^2(u) - \operatorname{sn}^2(u) \operatorname{dn}^2(u)}{1 - k^2 \operatorname{sn}^4(u)}, \quad \operatorname{dn}(2u) = \frac{\operatorname{dn}^2(u) - k^2 \operatorname{sn}^2(u) \operatorname{cn}^2(u)}{1 - k^2 \operatorname{sn}^4(u)},$$

together with the definition of $\tilde{v}_2$ and (14.3), the expression (14.18) simplifies to

$$\frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{(\lambda_1^2 - \lambda_2^2 - \lambda_3^2) k_1^{(1)'} - (\lambda_1^2 - \lambda_2^2 + \lambda_3^2)}{k_1^{(1)} (\lambda_1^2 + \lambda_2^2 - \lambda_3^2)}.$$

An analogous computation starting from $\nu^{(1)}$ yields the same value for

$$\frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})},$$

which proves the claim. □

By Lemma 1, (14.10) can be rewritten as

$$(14.20) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} - \frac{\frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} - \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{-\frac{\vartheta_2(2\tilde{v}_1, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_1, 2\tau_2^{(1)})} + \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

From this expression, one deduces that the poles of $u(x, t)$ occur at

$$2i\kappa^{(1)}(x + vt) = 2\tilde{v}_1 + 4in\pi + 4im\pi\tau_2^{(1)}, \quad n, m \in \mathbb{Z},$$

while the zeros occur at

$$2i\kappa^{(1)}(x + vt) = 2\tilde{v}_2 + 4in\pi + 4im\pi\tau_2^{(1)}, \quad n, m \in \mathbb{Z}.$$